**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)**
(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
**Accredited by NAAC with "A+" Grade.**
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

| **Regulation:R26** | **I/IV-B.Tech. I -Semester** |
|---|---|

**COMPUTER SCIENCE AND DESIGN**

**COURSE STRUCTURE**

**(With effect from 2026-27 admitted Batch onwards)**

| Course Code | Course Name | Category | L | T | P | Cr | C.I.E. | S.E.E. | Total Marks |
|---|---|---|---|---|---|---|---|---|---|
| A2610001 | English for Technical Communication | **HS** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2610002 | Linear Algebra and Calculus | **BS** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2610004 | Physics for advanced Computing | **BS** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2610005 | Computational Thinking and Problem Solving using C | **ES** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2610010 | Fundamentals of Digital Logic | **ES** | 2 | 0 | 0 | 2 | 40 | 60 | 100 |
| A2610009 | AI Tools and Applications | **ES** | 0 | 1 | 3 | 2.5 | 50 | 50 | 100 |
| A2610012 | Physics for Advanced Computing Lab | **BS** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |
| A2610013 | Computational Thinking and Problem Solving using C Lab | **ES** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |
| A2610015 | IT Workshop | **ES** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |
| A2610018 | NSS, Sports & Yoga | **MC** | 0 | 0 | 2 | 0 | 50 | | 50 |
| | | **TOTAL** | **14** | **1** | **14** | **21** | **450** | **500** | **950** |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2610001 | HS | 3 | -- | -- | 3 | 40 | 60 | 3 hrs |

## ENGLISH FOR TECHNICAL COMMUNICATION

(Common to all Branches)

**Course Objectives:**

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|---|---|
| 1. | Enable students identify the context and thematic information from social, academic, and workplace dialogues and texts. |
| 2. | Develop students' ability to use appropriate grammatical structures and vocabulary. |
| 3. | Equip students with contextual vocabulary for writing. |
| 4. | Familiarize students with reading strategies to comprehend and interpret texts. |
| 5. | Develop students' ability to demonstrate appropriate social media etiquette. |

**Course Outcomes**

| Sl. No | Outcome | Knowledge Level |
|---|---|---|
| 1. | Identify the thematic information from transactional dialogues and texts to construct coherent written and spoken discourses. | K2 |
| 2. | Apply appropriate grammatical structures and vocabulary to summarize texts on the related themes. | K3 |
| 3. | Apply contextual vocabulary to paraphrase texts and write letters. | K3 |
| 4. | Develop reading strategies to comprehend and interpret various texts for a global understanding. | K3 |
| 5. | Construct book/film reviews and demonstrate appropriate social media etiquette in writing, employing sentence synthesis techniques. | K3 |

**SYLLABUS**

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|---|---|
| **UNIT-I (10Hrs)** | **Effective Communication: Exploring Channels and Methods**<br>**Vocabulary:** Roots, Prefixes, and Suffixes; Homonyms<br>**Grammar:** Articles, Prepositions, Tenses.<br>**Listening:** Listening for main ideas and supporting details, Note-taking<br>**Speaking:** Introducing self and others, Describing people and objects, JAM<br>**Reading:** Definition and Importance, Skills and Sub-Skills of Reading<br>**Writing:** Cohesion and Coherence, Paragraph Writing |
| **UNIT-II (10 Hrs)** | **Eco-Consciousness: Negotiating for a Sustainable Future**<br>**Vocabulary:** Contractions, Phrasal Verbs, Idioms & Phrases<br>**Grammar:** Subject-Verb Agreement, Conditional Sentences<br>**Listening:** Listening to various texts and comprehension<br>**Speaking:** Role-plays, Extempore<br>**Reading:** Reading for Inference, Point of View, Tone and Perspective<br>**Writing:** Note-making, Summarizing, Paraphrasing (at basic level) |

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| **UNIT-III (10 Hrs)** | **Disruptive Technologies: Ethical Implications and Future Prospects**<br>**Vocabulary:** One-Word Substitutes, Commonly Confused Words<br>**Grammar:** Degrees of Comparison, Active and Passive Voice, Question Tags<br>**Listening:** Listening to stories, news, documentaries, and discussions<br>**Speaking:** Debate, Group Discussions<br>**Reading:** Reading the extracts/articles/ essays of the newspapers, journals, and literary texts.<br>**Writing:** Paraphrasing (at advanced level), Letter Writing |
| | |
| **UNIT-IV (10 Hrs)** | **Entrepreneurship and Start-Up Culture: Emerging Trends**<br>**Vocabulary:** Business/ Profession/ Career/ Job-related Vocabulary<br>**Grammar:** Sentence Synthesis, Simple, Compound and Complex Sentences<br>**Listening:** Listening to debates and discussions<br>**Speaking:** PowerPoint Presentations by Individuals and Teams<br>**Reading:** Reading general texts for global understanding<br>**Writing:** Email Writing, Data Interpretation, Report Writing |
| | |
| **UNIT-V (10 Hrs)** | **Ethical Leadership: Integrating Social Responsibility and Core Values**<br>**Vocabulary:** Advanced Vocabulary for GRE/TOEFL/ IELTS/CAT<br>**Grammar:** Sentence Synthesis, Para Jumbles<br>**Listening:** Listening to talks/speeches by eminent personalities<br>**Speaking:** Team Discussions and Presentations<br>**Reading:** Reading specific texts for global comprehension<br>**Writing:** Book/Film Reviews, Social Media Etiquette |
| | |
| **Textbook:** | |
| 1. | Chekuri, T., & Perumalla, R. K. (2026). NextGen English for engineers: A course book for technical English communication. Cengage. |
| **Reference Books:** | |
| 1. | Technical Communication (2nd Ed.) by M. Raman & S. Sharma. Oxford University Press, 2012. |
| 2. | Effective Technical Communication by M.A. Rizvi, Tata McGraw-Hill, 2005. |
| 3. | English for Engineers and Technologists, Orient Blackswan 2010 (Combined edition, Vol.1 &2). |
| 4. | The Oxford Guide to Writing and Speaking by John Seely, Oxford University Press, 2004. |
| | |
| **e-Resources** | |
| 1. | eGyanKosh. (n.d.). National digital repository. https://egyankosh.ac.in/ |
| 2. | Read Theory. (n.d). Adaptive reading comprehension practice. https://readtheory.org/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2610002 | BS | 3 | -- | -- | 3 | 40 | 60 | 3 Hrs. |

## LINEAR ALGEBRA & CALCULUS

**(Common to All Branches of Engineering)**

**Course Objectives:** Students are expected to learn

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|---|---|
| 1. | Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations. |
| 2. | Eigen values, Eigen vectors and quadratic forms. |
| 3. | Problems and applications of Mean value theorems |
| 4. | Application of partial differentiation for determining maxima/ minima of functions. |
| 5. | Concepts of double, triple integrals and its applications. |

**Course Outcomes:**After completion of the course, the student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | **Solve** a system of linear equations by using direct and indirect methods. | K3 |
| 2. | **Determine** Eigen values and Eigen vectors, execute matrix Diagonalization and reduce Q.F.'s to canonical forms. | K3 |
| 3. | **Apply** mean value theorems to analyze function behaviour and solve real life problems. | K3 |
| 4. | **Apply** the concepts of partial differentiation in various engineering applications. | K3 |
| 5. | **Apply** the concepts of multiple integrals to calculate areas, volumes of geometrical regions and in engineering applications. | K3 |

## SYLLABUS

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|---|---|
| **UNIT-I (10Hrs)** | **Linear System of Equations:** Rank of a matrix by echelon form, normal form. Inverse of a matrix by Gauss-Jordan method. Consistency of linear equations: Homogeneous and Non-Homogeneous systems. Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods. |
| **UNIT-II (10 Hrs)** | **Eigen Values, Eigen Vectors and Quadratic Forms:** Eigen values and Eigen vectors of a matrix, their properties, Cayley-Hamilton Theorem (without proof), Inverse and powers of a matrix by Cayley – Hamilton theorem, Diagonalization of a matrix, Quadratic forms and Nature of the Quadratic Forms, Reduction of a Quadratic form to canonical form by Orthogonal Transformation and Singular Value Decomposition. |
| **UNIT-III (10 Hrs)** | **Single Variable Calculus:** Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem, Geometrical interpretation of these theorems, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems (without proof) with Lagrange's form of remainder, Taylor's and Maclaurin series. |

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|---|---|
| **UNIT-IV (10 Hrs)** | **Partial differentiation and Applications:** Functions of several variables, Continuity and Differentiability, Partial derivatives, Euler's theorem, Total derivative, chain rule, Taylor's and Maclaurin's series expansion of function of two variables, Jacobians, Functional dependence, Maxima and Minima of function of two variables without constraints and with constraints - Method of Lagrange multipliers. |
| | |
| **UNIT-V (10 Hrs)** | **Multiple Integrals:** Double integrals, change of order of integration, triple integrals, change of variables from Cartesian to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals). |
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| **Textbooks:** | |
|---|---|
| 1. | HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition |
| 2. | AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition. |
| **Reference Books:** | |
| 1. | ThomasCalculus,GeorgeB.Thomas,MauriceD.WeirandJoelHass,PearsonPublishers,2018, 14thEdition. |
| 2. | Higher Engineering Mathematics, B.V.Ramana, McGraw HillEducation,2017. |
| 3. | Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition. |
| 4. | Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th edition. |
| 5. | Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications,2014, Third Edition (Reprint 2021). |
| 6. | Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty Two EditionReprint 2022). |
| 7. | Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015. |

| **e-Resources** | |
|---|---|
| 1. | https://en.wikipedia.org/wiki/System_of_linear_equations |
| 2. | https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors |
| 3. | https://www.math.hmc.edu/calculus/tutorials/eigenstuff/ |
| 4. | https://en.wikipedia.org/wiki/Quadratic_form |
| 5. | https://en.wikipedia.org/wiki/Calculus |
| 6. | https://en.wikipedia.org/wiki/Partial_derivative |
| 7. | https://www.whitman.edu/mathematics/calculus_online/section14.03.html |
| 8. | https://en.wikipedia.org/wiki/Multiple_integral |
| 9. | http://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2610004** | **BS** | **3** | **--** | **--** | **3** | **40** | **60** | **3 Hrs.** |

**PHYSICS FOR ADVANCED COMPUTING**

(**Common to CSBS, CSD, CSIT**)

**Course Objectives:**

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|---|---|
| 1. | To equip students the concepts of interference, diffraction and polarization and their applications in optics and communication systems. |
| 2. | To understand the basic concepts of Laser light sources and working principle of optical fibers and identify the engineering applications of lasers and optical fibers in communication. |
| 3. | To explain various electron theories and interpret the effect of temperature on Fermi-Dirac distribution function and provide basic knowledge of quantum mechanics for quantum computing applications. |
| 4. | To familiarize students with semiconductors and magnetic materials and their applications. |
| 5. | To impart knowledge on low temperature physics and superconductivity and their applications. |

**Course Outcomes:** After successful completion of this course, the student will be able to:

| S.No. | Outcome | Knowledge Level |
|---|---|---|
| 1. | Apply the principles of interference, diffraction and polarization to analyze optical systems and engineering applications. | K3 |
| 2. | Analyze the operation of various laser systems as well as the working principle of optical fibers in relation to communication applications. | K4 |
| 3. | Interpret the effects of temperature on Fermi energy and Apply the concepts of quantum mechanics to explain modern computational systems like Quantum Computing. | K3 |
| 4. | Analyse semiconductor characteristics and magnetic material behaviour for engineering and electronic applications. | K4 |
| 5. | Apply the concepts of low-temperature physics and superconductivity in modern technological applications and intelligent systems. | K3 |

**SYLLABUS**

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|---|---|
| **UNIT-I (10Hrs)** | **APPLIED WAVE OPTICS**<br>**Interference:** Superposition Principle – Coherence – Types of Interference – Thin-film interference (reflection geometry) – Newton's rings – Determination of wavelength and refractive index.<br>**Diffraction and Optical Surveying:** Fresnel and Fraunhofer diffraction – Fraunhofer's diffraction due to single slit (quantitative), double slit & N-slits (qualitative) – Diffraction Grating and grating spectrum – Dispersive and Resolving power of a grating (qualitative).<br>**Polarization:** Types of polarization – Nicol's prism – Half-wave and Quarter-wave |

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| | plates. |
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| **UNIT-II (9 Hrs)** | **QUANTUM PHOTONICS**<br>**Lasers:** Principles of Absorption – Spontaneous and Stimulated Emission – Population Inversion – Three level and Four level laser systems – Construction and working of Ruby and He-Ne Laser – Applications (qualitative).<br>**Fiber Optics:** Propagation mechanism – Numerical Aperture – Acceptance angle – Types of optical fibres – Classifications and Modes of optical fibres. |
| | |
| **UNIT-III (10 Hrs)** | **FREE ELECTRON THEORY & QUANTUM MECHANICS**<br>**Free Electron Theory:** Classical free electron theory – merits and demerits – Quantum free electron theory – equation for electrical conductivity – Fermi-Dirac statistics – Fermi level and variation with temperature.<br>**Quantum Mechanics:** de Broglie concept of matter waves – Heisenberg's uncertainty principle – Wave function – Schrödinger's time dependent and time-independent wave equations – Particle in a one-dimensional potential well – Quantum tunneling (basic idea). |
| | |
| **UNIT-IV (10 Hrs)** | **SEMICONDUCTORS & MAGNETIC MATERIALS**<br>**Semiconductors:** Formation of energy bands – Classification of solids - Intrinsic semiconductors – Density of charge carriers – Electrical conductivity – Fermi level – extrinsic semiconductors: density of charge carriers (qualitative) – dependence of Fermi energy on carrier concentration and temperature – Drift and diffusion currents – Einstein's equation – Hall effect – Hall coefficient – Applications of Hall effect.<br>**Magnetic Materials:** Introduction and classification - Dia, Para, Ferro, Ferri, and Anti-Ferro; Hysteresis and Eddy current losses in electrical machines – Skin effect; Soft & Hard magnetic materials. |
| | |
| **UNIT-V (9 Hrs)** | **LOW TEMPERATURE PHYSICS & SUPERCONDUCTIVITY**<br>**Low Temperature Physics:** Introduction – Low temperature and its importance – Properties of materials at low temperatures – Liquefaction of gases (basic concept only) – Liquid Nitrogen and liquid Helium (basic properties and uses).<br>Superconductors: Properties – Critical parameters – Meissner effect – Type-I and Type-II superconductors – BCS theory – Cooper pairs and energy gap – DC and AC Josephson effect – Applications (qualitative only): SQUIDs, magnetic levitation and lossless power transmission. |
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| **Textbooks:** | |
| 1. | Avadhanulu, M.N., Kshirsagar, P.G., & Murthy, T.V.S.A., A textbook of Engineering Physics, S. Chand Publishing (2019) |
| 2. | P.K. PalaniSamy, Engineering Physics, SciTech Publications (2019) |
| **Reference Books:** | |

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|---|---|
| 1. | D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015 |
| 2. | B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2021 |
| 3. | S. O. Pillai, Solid State Physics, New Age International (2019) |
| | |
| **e-Resources** | |
| 1. | NPTEL – nptel.ac.in |
| 2. | MIT OpenCourseWare – ocw.mit.edu |
| 3. | HyperPhysics – hyperphysics.phy-astr.gsu.edu |
| 4. | University of Colorado Boulder. (n.d.). PhET interactive simulations. https://phet.colorado.edu/ |
| 5. | Library of Congress. (n.d.). Science and technology: Selected internet resources – Physics. https://www.loc.gov/rr/scitech/selected-internet/physics.html |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2610005** | **ES** | **3** | **--** | **--** | **3** | **40** | **60** | **3 Hrs** |

**COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C**

(Common to AIML, CIC, CSD, CSE, CSIT, ECE, IT)

**Pre-requisite**: Mathematics and logical reasoning at 10+2 level. No prior programming experience is assumed.

**Course Objectives:** The main objectives of the course are to

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|---|---|
| 1. | Familiarize students with computers, computational thinking, and the process of algorithmic problem solving. |
| 2. | Build strong programming fundamentals using the C language, including operators and control structures. |
| 3. | Develop the ability to design and implement solutions using arrays, strings, and user- defined data types. |
| 4. | Impart working knowledge of pointers, recursion, dynamic memory allocation, functions, and file handling for building modular, reusable programs. |

**Course Outcomes:** At the end of the course, student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | Apply computational thinking strategies to construct algorithms, flowcharts, and implement them as C programs. | K3 |
| 2. | Apply selection and iterative control structures along with arrays to solve programming problems. | K3 |
| 3. | Analyze problems involving strings and user-defined datatypes to construct suitable C solutions. | K4 |
| 4. | Apply pointers, dynamic memory allocation, and recursive techniques to design modular C programs using functions. | K3 |
| 5. | Apply preprocessor directives and file handling techniques to perform file input-output operations in C programs. | K3 |

**SYLLABUS**

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|---|---|
| **UNIT-I (10Hrs)** | **Computational Thinking and Introduction to C**<br>**Computational Thinking:** Decomposition, Pattern Recognition, Abstraction, Algorithm Design; representing solutions as algorithms, pseudocode, and flowcharts; Number Systems and Data Representation (integers, reals); Overview of Computer Organization and Program Execution.<br>**Introduction to C:** Background, Structure of a C Program, Identifiers, Keywords, Variables, Constants, Basic Data Types, Input/Output; Expressions, Operators, type |

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|---|---|
| | of operators- arithmetic, logical, relational, assignment, conditional, increment, decrement, etc., unary, binary, ternary, Operator Precedence and Associativity, Type Conversion; Scope, Storage Classes, and Type Qualifiers.<br>**Bitwise Operators:** Logical Bitwise Operators, Shift Operators. |
| | |
| **UNIT-II (10 Hrs)** | **Selection and Decision-Making**: Two-Way and Multi-Way Selection (if-else, switch), Nested Conditionals.<br>**Repetition:** Pre-test and Post-test Loops, Counter- and Event-Controlled Loops, for/while/do-while, break and continue, Nested Loops; Programming Applications combining Selection and Repetition.<br>**Arrays:** One-Dimensional and Two-Dimensional Arrays, Multidimensional Arrays, Array-based Problem Solving (searching, sorting, averaging). |
| | |
| **UNIT-III (10 Hrs)** | **Strings:** String Representation, Standard String-Handling Functions, Arrays of Strings, String-to-Data Conversion.<br>**User-Defined Types:** typedef, Enumerated Types, Structures, Unions, and Nested Structures; Programming Applications using Structures and Arrays of Structures |
| | |
| **UNIT-IV (10 Hrs)** | **Pointers:** Pointer Fundamentals, Pointer-to-Pointer, Pointer Arithmetic, Pointers and Arrays, Array of Pointers, Compatibility and l-value/r-value concepts. Dynamic Memory Allocation: malloc, calloc, realloc, free.<br>**Functions:** Designing Structured/Modular Programs, Function Declaration and Definition, Parameter Passing (by value, by reference using pointers), Passing Arrays and Pointers to Functions, Standard Library Functions, Inter-Function Communication.<br>**Recursion:** Recursive vs. Iterative Solutions, Designing Recursive Algorithms, Recursion with Arrays and Pointers, Efficiency Considerations. |
| | |
| **UNIT-V (10 Hrs)** | **Preprocessor**: Directives and Macros.<br>**File I/O:** Streams, Text vs. Binary Files, Standard Library File Functions, formatted and Character I/O, Sequential File Processing. |
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| **Textbooks:** | |
| 1. | Behrouz A. Forouzan & Richard F. Gilberg, Computer Science: A Structured Programming Approach Using C, 3rd Edition, Cengage Learning, ISBN-13: 978-0-534-49132-1. |
| 2. | Brian W. Kernighan & Dennis M. Ritchie, The C Programming Language, 2nd Edition, Pearson Education / Prentice Hall India, 1988. |
| **Reference Books:** | |
| 1. | Sumitabha Das, "Computer Fundamentals and Programming", 18th Edition, McGraw Hill Education, 2018 |
| 2. | Ashok N. Kamthane, Amit Ashok Kamthane, "Programming in C", 4th Edition, Pearson Education, 2025. ISBN: 978-9332543633. |
| 3. | Pradip Dey, Manas Ghosh, "Computer Fundamentals and Programming in C", 2nd Edition, |

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|---|---|
| | Oxford University Press, 2013. ISBN: 978-0198068842 |
| | |
| **e-Resources** | |
| 1. | NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview |
| 2. | Geeks for Geeks – C Programming Language: https://www.geeksforgeeks.org/c-programming-language/ |
| 3. | Programiz – Learn C Programming: https://www.programiz.com/c-programming |
| 4. | cplusplus.com – C Standard Library Reference: https://cplusplus.com/reference/clibrary/ |
| 5. | TutorialsPoint – C Programming: https://www.tutorialspoint.com/cprogramming/index.html |
| 6. | W3Schools – C Tutorial: https://www.w3schools.com/c/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2620010** | **ES** | **2** | **--** | **--** | **2** | **40** | **60** | **3Hrs** |

**FUNDAMENTALS OF DIGITAL LOGIC**

**(Common for CSD & CSIT)**

**Pre-requisite**: Basic Algebra, Set Theory Concepts, Binary Concepts and Logical Reasoning

**Course Objectives:** The main objectives of the course are to

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|---|---|
| 1 | Develop proficiency in number systems, codes, and Boolean algebra as the mathematical basis of digital logic. |
| 2 | Enable students to analyze and design combinational logic circuits using minimization techniques and standard building blocks. |
| 3 | Introduce sequential logic elements such as latches and flip-flops and their role in storing digital state. |
| 4 | Build the ability to design synchronous sequential circuits, including registers, counters, and finite state machines. |

**Course Outcomes** :At the end of the course, student will be able to

| S.No | Outcomes | Knowledge Level |
|---|---|---|
| 1 | Explain number systems, binary arithmetic, complements, binary codes, Boolean algebra, and the operation of basic and universal logic gates | K2 |
| 2 | Apply Boolean algebra and Karnaugh-map techniques to simplify Boolean expressions and design basic combinational circuits such as adders and subtractors. | K3 |
| 3 | Analyze and design combinational logic circuits using multiplexers, demultiplexers, encoders, decoders, and magnitude comparators. | K4 |
| 4 | Explain and analyze latches and flip-flops, including SR, D, JK, and T flip-flops, using excitation tables and timing diagrams. | K4 |
| 5 | Design and evaluate sequential circuits using shift registers and asynchronous, synchronous, up/down, and modulo-N counters | K5 |

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| **UNIT – 1 (7 Hrs)** | Number Systems and Boolean Algebra Number systems (binary, octal, decimal, hexadecimal) and inter-conversions; binary arithmetic; complements (1's and 2's); binary codes (BCD, Gray code, ASCII); Boolean algebra postulates and theorems; logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) and universal gates |
| **UNIT-II (5 Hrs)** | Simplification of Boolean functions<br>Minimization using Karnaugh maps (up to 4 variables); Sum-of-Products and |

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|---|---|
| | Product-of-Sums forms;<br>Combinational Logic Design I<br>design of half-adder, full-adder, half-subtractor, full-subtractor |
| | |
| **UNIT – 3**<br>**(5 Hrs)** | Combinational Logic Design II:<br>multiplexers and demultiplexers; encoders and decoders; magnitude comparators.<br>Sequential Logic Fundamentals- Latches and FlipFlops, Difference between combinational and sequential circuits |
| | |
| **UNIT – 4**<br>**(5 Hrs)** | Sequential Logic Design—-SR latch; clocked SR, D, JK, and T flip-flops; masterslave flip-flop; flip-flop excitation tables; timing diagrams; setup and hold time. |
| | |
| **UNIT – 5**<br>**(5 Hrs)** | Registers, Counters, and Sequential Circuit Design<br>Registers, Shift registers (SISO, SIPO, PISO, PIPO); ripple (asynchronous) counters; synchronous counters; up/down and modulo-N counters |
| | |
| **Text Books**: | |
| 1 | Mano, M. M., Ciletti, M. D. (2018). Digital design (6th ed.). Pearson. |
| 2 | Floyd, T. L. (2015). Digital fundamentals (11th ed.). Pearson. |
| **Reference Books:** | |
| 1 | Hill, F. J., & Peterson, G. R. (2010). Introduction to switching theory and logic design (3rd ed.). Wiley. |
| 2 | Wakerly, J. F. (2018). Digital design: Principles and practices (5th ed.). Pearson. |
| 3 | Brown, S., &Vranesic, Z. (2014). Fundamentals of digital logic with VHDL design (3rd ed.). McGraw-Hill Education. |
| **e References:** | |
| 1 | NPTEL. (n.d.). Digital circuits and systems. National Programme on Technology Enhanced Learning. https://nptel.ac.in/ |
| 2 | MIT OpenCourseWare. (n.d.). Digital systems laboratory. Massachusetts Institute of Technology. https://ocw.mit.edu/courses/6-111-introductory-digital-systems-laboratory-fall-2019/ |
| 3 | All About Circuits. (n.d.). Digital circuits. https://www.allaboutcircuits.com/textbook/digital/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2610009** | **ES** | **--** | **1** | **3** | **2.5** | **50** | **50** | **3Hrs** |

**AI TOOLS & APPLICATIONS**

(Common to AIDS, CIVIL, CSBS, CSD, CSIT, EEE, IT, MECH)

**Course Objectives:** To familiarize students with the landscape of AI tools and their underlying working principles.

| | |
|---|---|
| 1 | To develop competence in prompt engineering for effective interaction with generative AI systems |
| 2 | To enable students to apply AI tools for data analysis, visualization, and interpretation. |
| 3 | To train students in using AI-assisted tools for coding, writing, and productivity tasks. |
| 4 | To instill awareness of ethical, legal, and societal considerations in AI tool usage. |

**Course Outcomes:**

| S. No | Outcome | Knowledge Level |
|---|---|---|
| 1 | Illustrate the categories, capabilities, and limitations of major AI tools (text, image, code, data). | K3 |
| 2 | Apply the principles of prompt engineering and demonstrate how generative AI models process instructions. | K3 |
| 3 | Apply AI tools to solve first-year engineering problems in analysis, writing, and coding tasks. | K3 |
| 4 | Analyze AI-generated outputs for accuracy, bias, and reliability before use. | K4 |
| 5 | Evaluate the ethical, academic-integrity, and societal implications of AI tool use, justifying appropriate usage choices. | K4 |

**SYLLABUS**

**1**

Introduction to AI and the AI Tools Landscape What is AI, Machine Learning, and Generative AI — basic definitions and how they relate to one another. Core categories of AI tools: conversational/text, code assistants, data-analysis tools, image-generation tools, and writing/productivity tools. How large language models process and generate responses (conceptual, non-mathematical overview). Criteria for selecting the right AI tool for a given task — accuracy needs, cost, data sensitivity, and output format. Setting up accounts and workspaces; understanding free-tier versus paid-tier capabilities and limitations.

**Widely Used AI Tools by Application (for classroom coverage):**

| Category | Example Tools | Typical Application |
|---|---|---|
| Conversational/General-Purpose AI | ChatGPT, Claude, Gemini, Perplexity | Writing, research, coding help, file/data analysis, cited web search |
| AI Coding Assistants | GitHub Copilot, Cursor, Claude Code | In - IDE code completion, AI Native editing, multi-file |

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| | | agentic coding |
| Data Analysis Tools | ChatGPT(Advanced Data Analysis), Microsoft Copilotin Excel, Julius AI | Upload a dataset, generate statistics/charts, spreadsheet Formulas via natural language |
| Image Generation | Mid journey, DALL·E 3, Stable Diffusion/Flux | Illustrations, concept art, design mockups |
| Writing & Productivity | Grammarly, Gamma, Fathom, Notion AI | Grammar/style checking, slide generation, meeting summaries, task automation |
| Audio, Video & Presentation (optional) | Eleven Labs, Runway/Sora, Synthesia | Text-to-speech, AI video generation, script-to-video presentations |

**Teacher Practical Examples (for classroom demonstration):**

**1.** Live comparison of the same general engineering query (e.g., "explain the engineering design process") answered by two different AI chatbots, highlighting differences in response style and accuracy.

**2.** Walkthrough of an AI tool's interface — settings, conversation history, file upload, and model selection — using a common platform.

**Student Exercise Problems (for practice):**

**1.** Create accounts on two AI platforms and compare their responses to a simple engineering question (e.g., "explain the difference between analysis and design").

**2.** Prepare a one-page comparison chart of five AI tools by category and primary use case.

**3.** Write a short reflection identifying one task in your daily academic life where an AI tool could help.

2

**Prompt Engineering and Generative AI Tools**

Anatomy of a good prompt — context, clear instruction, desired format, and constraints.

Zero-shot, few-shot, and chain-of-thought prompting techniques and when to use each.

Role-based and context-setting prompts (e.g., asking the AI to respond as a reviewer, tutor, or domain expert).

Iterative prompt refinement — diagnosing a poor AI response and rewriting the prompt to fix it.

Using generative AI for brainstorming, summarizing, and drafting technical content.

**Teacher Practical Examples (for classroom demonstration):**

**1.** Refine a vague prompt (e.g., "explain project scheduling") into a precise one step-by-step, showing output improvement at each stage.

**2.** Demonstrate few-shot prompting by providing examples to guide the AI toward a specific output format (e.g., a formatted comparison table).

**Student Exercise Problems (for practice):**

**1.** Write three versions of a prompt (basic, detailed, role-based) for the same task and compare outputs.

**2.** Use an AI tool to summarize a one-page technical article and verify the summary against the original.

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| | **3.** Draft a chain-of-thought prompt to solve a multi-step numerical/logical problem (e.g., a unit-conversion or estimation problem) and check the reasoning. |
| 3 | **AI Tools for Data Analysis and Visualization**<br>AI-assisted spreadsheet and dataset analysis using natural-language commands.<br>Natural-language querying of datasets to extract summary statistics and insights.<br>AI tools for generating charts, graphs, and visual summaries from tabular data.<br>Interpreting AI-generated statistical summaries correctly and recognizing their limitations.<br>Identifying and correcting errors or misinterpretations in AI-generated data analysis.<br>**Teacher Practical Examples (for classroom demonstration):**<br>**1.** Upload a small dataset (e.g., class attendance or survey data) to an AI tool and generate a summary-statistics report and chart using natural-language commands.<br>**2.** Show an example where an AI tool misinterprets data (e.g., treats a numeric code as text), and demonstrate how to verify and correct it.<br>**Student Exercise Problems (for practice):**<br>**1.** Use an AI tool to clean and summarize a small sample dataset (e.g., class attendance or marks data).<br>**2.** Generate three different chart types from the same dataset using AI prompts and justify which is most appropriate.<br>**3.** Cross-check one AI-generated statistical claim manually and report any discrepancy found. |
| 4 | **AI Tools for Coding, Writing, and Productivity**<br>AI coding assistants — code generation, explanation, and debugging support for simple programs.<br>AI writing assistants for reports, emails, and technical documentation.<br>AI tools for scheduling, note-taking, and meeting/lecture summarization.<br>AI-assisted presentation and slide-outline generation from a topic prompt.<br>Integrating AI tools into an efficient day-to-day academic and engineering workflow.<br>**Teacher Practical Examples (for classroom demonstration):**<br>**1.** Use an AI coding assistant to generate a simple program (e.g., a temperature converter), then intentionally introduce a bug and have the AI debug it.<br>**2.** Generate a presentation outline and speaker notes from a topic prompt using an AI writing/productivity tool.<br>**Student Exercise Problems (for practice):**<br>**1.** Use an AI coding tool to write and debug a simple program for a given problem statement.<br>**2.** Draft a formal email or short report using an AI writing assistant, then manually edit it for accuracy and tone.<br>**3.** Generate a slide outline for a 5-minute seminar topic using an AI tool and refine it. |
| 5 | **Ethics, Reliability, and Emerging Trends**<br>Bias and hallucination in AI outputs, and techniques for detecting and fact-checking them.<br>Academic integrity and appropriate citation of AI assistance in coursework and submissions.<br>Data privacy and security considerations, including obligations under India's Digital Personal Data Protection Act, 2023 (DPDP Act), when uploading personal or institutional data.<br>Copyright and intellectual property considerations in AI-generated text and image content. |

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| | Emerging trends — multimodal AI and AI agents — and their implications for future engineering practice.<br>**Teacher Practical Examples (for classroom demonstration):**<br>**1.** Show a documented case of AI hallucination (a fabricated fact or citation) and demonstrate how to fact-check it.<br>**2.** Walk through an example of properly citing AI assistance in an academic submission per institutional guidelines.<br>**3.** Walk through a worked example of checking whether AI-generated text or image output raises a copyright/IP concern (e.g., close paraphrasing of a source).<br>**Student Exercise Problems (for practice):**<br>**1.** Identify and document a hallucination or bias instance produced by an AI tool on a factual query.<br>**2**. Draft an "AI usage declaration" for a hypothetical assignment, citing tools and extent of use appropriately.<br>**3.** Review the data-handling/privacy policy of one AI tool used in this course and identify one DPDP Act, 2023 compliance consideration relevant to student data. |
| | |
| **Text Books:** | |
| 1 | Marr, B. (2024). Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society. Wiley. |
| 2 | Mollick, E. (2024). Co-intelligence: Living and working with AI. Portfolio/Penguin. |
| **Reference Books:** | |
| 1 | Selwyn, N. (2022). Education and technology: Key issues and debates (3rd ed.). Bloomsbury. |
| 2 | Susskind, R., & Susskind, D. (2022). The future of the professions: How technology will transform the work of human experts (Updated ed.). Oxford University Press. |
| 3 | Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson. |
| **Web Resources:** | |
| 1 | Google. (n.d.). Google AI essentials. Grow with Google. https://grow.google/ai-essentials/ |
| 2 | OpenAI. (n.d.). OpenAI documentation and guides. https://platform.openai.com/docs |
| 3 | Anthropic. (n.d.). Prompt engineering overview. Claude Docs. https://docs.claude.com/en/docs/build-with-claude/prompt- Engineering/overview |
| 4 | UNESCO. (2023). Guidance for generative AI in education and research. https://unesdoc.unesco.org/ |
| 5 | AICTE. (2025). AICTE inputs on Artificial Intelligence — Year of AI initiative. https://www.aicte.gov.in/downloads/initiatives/AICTE%20inputs%20on%20Artificial%20Intelligence%20-compre hensive.pdf |
| 6 | Ministry of Electronics and Information Technology, Government of India. (2023). Digital Personal Data Protection Act, 2023. https://www.meity.gov.in/content/digital-personal-data-protection-act-2023 |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2610012** | **BS** | **--** | **--** | **3** | **1.5** | **50** | **50** | **3 Hrs.** |

**PHYSICS FOR ADVANCED COMPUTING LAB**

(**Common to CSBS, CSD, CSIT**)

**Course Objectives:**

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| 1 | To impart hands on experience to the students entering Engineering & Technology education about handling equipment/instruments and use them in experimentation. |
| 2 | To make the students understand theoretical aspects of various phenomena experimentally. |

**Course Outcomes:**

| S.No. | Outcome | Knowledge Level |
|---|---|---|
| 1 | Apply fundamental principles of Physics to conduct engineering laboratory experiments. | K3 |
| 2 | Get hands on experience in setting up experiments and using the instruments/equipment individually and conduct experiments. | K3 |
| 3 | Apply graphical and analytical methods to analyze experimental data and determine physical parameters. | K3 |
| 4 | Get introduced to using new/advanced technologies and understand their significance . | K3 |

**List of Experiments**

(Any **10** of the following listed experiments)

(Out of which any **TWO** experiments may be conducted in virtual mode)

| | |
|---|---|
| 1 | Determination of thickness of thin object by wedge method. |
| 2 | Determination of radius of curvature of a given plano convex lens by forming Newton's rings. |
| 3 | Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration. |
| 4 | Determination of resolving power of telescope. |
| 5 | Determination of dielectric constant of given capacitor by charging and discharging method. |
| 6 | Study the variation of B versus H by magnetizing the magnetic material (B-H curve). |
| 7 | Determination of temperature coefficients of a thermistor. |
| 8 | Determination of wavelength of Laser and grating element. |
| 9 | Estimation of Planck's constant using photoelectric effect. |
| 10 | Determination of the Resistivity of semiconductor by four probe method. |
| 11 | Determination of the energy gap of a semiconductor using p-n junction diode. |
| 12 | Study the variation of magnetic field along the axis of a current carrying circular coil by Stewart & Gee's Method |
| 13 | Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall Effect. |
| 14 | Determination of Acceleration due to gravity and Radius of gyration using Compound pendulum. |

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| 15 | Determination of magnetic susceptibility by Kundt's tube method |
| 16 | Verification of Brewster's law. |
| 17 | Determination of frequency of A.C. supply using Sonometer. |
| 18 | Determination of frequency of electrically maintained tuning fork by Melde's apparatus. |
| 19 | Determination of Rigidity modulus of material of a wire-dynamic method (Torsional pendulum). |
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| **Reference Books:** | |
| 1 | S. Balasubramanian, M.N. Srinivasan "A Text Book of Practical Physics"- S Chand Publishers, 2017. |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2610013 | ES | -- | -- | 3 | 1.5 | 50 | 50 | 3 Hrs |

**COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C LAB**

(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, ECE, IT)

**Pre-requisite**: - Basic understanding of computer fundamentals and problem-solving concepts

**Course Objectives:** The main objectives of the course are to

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| 1 | Apply the principles of C language and computational thinking techniques in problem solving. |
| 2 | Design flowcharts and algorithms, and to develop the ability to trace and debug programs. |
| 3 | Design and develop C programs using arrays, strings, pointers, recursion, and functions. |
| 4 | Apply dynamic memory allocation, file operations, and preprocessor commands in program design. |

**Course Outcomes:**

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1 | Apply computational thinking to construct algorithms and flowcharts for a given Problem and translate them into C programs. | K3 |
| 2 | Develop C programs using operators, selection statements and loops to implement a specified logic. | K3 |
| 3 | Develop C programs that manipulate arrays, strings, structures and pointers to solve computational problems. | K3 |
| 4 | Construct modular programs using user-defined functions and recursion and perform file input-output operations. | K3 |
| 5 | Analyze a C program by tracing and debugging it and evaluate alternative Implementations for correctness and efficiency. | K4 |

**SYLLABUS**

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|---|---|
| 1 | **Exercise 1: Getting Started with**<br>1. Write, compile and run a C program that prints "Hello, World!" on the screen. Identify the source file, the object file and the executable produced, and record the command used at each stage.<br>2. Write a C program that declares variables of type int, float, char and double, reads a value for each from the user, and displays each value together with the size of its type using size of.<br>3. Write a C program to read the height and width of a rectangle from the user and compute its perimeter and area. |
| 2 | **Exercise 2: Operators and Expressions**<br>1. Write a C program to compute the distance between two points whose coordinates are entered by the user.<br>2. Write a C program that reads four integers p, q, r, s and, using relational and logical operators, prints 'Correct values' if q > r, s > p and (r + s) > (p + q); otherwise 'Wrong values'.<br>3. Write a C program to compute simple and compound interest for values entered by the user. |

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| 3 | **Exercise 3: Selection and Decision Making**<br>1. Write a C program to read a character and classify it as a vowel, consonant, digit or special symbol.<br>2. Write a C program to find the largest of three numbers using nested if-else.<br>3. Write a menu-driven C program to compute the area of various geometrical shapes (circle, rectangle, triangle, square) using switch-case. |
| 4 | **Exercise 4: Loops and Series**<br>1. Write a C program to display the first n even natural numbers and their sum.<br>2. Write a C program to reverse a given integer and check whether it is a palindrome.<br>3. Write a C program to display the sum of the harmonic series 1 + 1/2 + 1/3 + ... up to n terms. |
| 5 | **Exercise 5: Nested Loops and Number Patterns**<br>1. Write a C program to print the multiplication table pyramid for numbers 1 to n using nested loops.<br>2. Write a C program to check whether a given number is prime, then extend it to print all prime numbers and all Armstrong numbers below. |
| 6 | **Exercise 6: One-Dimensional Arrays**<br>1. Write a C program to print all unique elements of an array and implement linear search to find a given element in the array.<br>2. Write a C program to separate the odd and even elements of an array into two different arrays and implement binary search to find a given element in a sorted array.<br>3. Write a C program to sort the elements of an array in ascending order using selection sort and compare the efficiency of linear search vs. binary search on the sorted array by counting the number of comparisons each method takes. |
| 7 | **Exercise 7: Matrices**<br>. Write a C program to perform addition of two square matrices and display the resultant matrices.<br>. Write a C program to multiply two square matrices and display the resultant matrices.<br>. Write a C program to find the transpose of a given matrix. |
| 8 | **Exercise 8: Strings:**<br>Write C programs to perform the following without using string handling functions<br>• **Find the length** of a given string.<br>• **Copy** one string into another.<br>• **Concatenate** two strings.<br>• **Compare** two strings and display whether they are equal, or which one is greater.<br>• **Reverse** a given string (using a separate swap function to exchange characters).<br>• **Search for a character** in a string and display its position (or indicate if not found).<br>• **Count the number of vowels, consonants, and words** in a given string |
| 9 | **Exercise 9: Pointers Fundamentals**<br>1. Write a C program to demonstrate the use of the & (address-of) and * (value-at-address) operators.<br>2. Write a C program to add two numbers using pointers.<br>3. Write a C program to swap two values using call by reference. |

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| 10 | **Exercise 10: Pointers and Dynamic Memory**<br>. Write a C program to dynamically allocate memory for an array using malloc, find the largest element and the sum of all elements in it, then repeat using calloc and note the difference between the two.<br>. Write a C program to count the number of vowels in a string using a pointer. |
| 11 | **Exercise 11: User-Defined Functions**<br>. Write a C program to check whether a number is prime using a user-defined function.<br>. Write a C program to compute the GCD and LCM of two numbers using functions.<br>. Write a C program to demonstrate a function that returns a pointer. |
| 12 | **Exercise 12: Recursion**<br>. Write a C program to compute the factorial and the n-th Fibonacci term using recursion.<br>. Write a C program to compute the sum of digits of a number using recursion.<br>. Write a C program to convert a decimal number to its binary equivalent using a recursive function. |
| 13 | **Exercise 13: File Handling**<br>. Write a C program to append multiple lines at the end of an existing text file.<br>. Write a C program to copy the contents of one file into another file.<br>. Write a C program to count the number of characters, words and lines in a text file. |
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| Reference Books: **Web References:** | |
| 1 | NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview |
| 2 | Virtual Labs – Problem Solving Lab, IIIT Hyderabad: https://ps-iiith.vlabs.ac.in/ |
| 3 | GeeksforGeeks – C Programming: https://www.geeksforgeeks.org/c-programming-language/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2610015** | **ES** | **--** | **--** | **3** | **1.5** | **50** | **50** | **3 Hrs** |

**IT Workshop**

(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, EEE, IT)

**Course Objectives:**

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| 1. | Understand the basic components of a computer, assemble and disassemble a PC, and install an operating system. |
| 2. | Enable students to connect to and navigate the Internet safely, and to understand netiquette and cyber hygiene practices. |
| 3. | Develop proficiency in office automation tools (Word, Excel, and PowerPoint) for academic and professional documentation, data analysis, and presentation. |

**Course Outcomes:** After completion of the course, the student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | Identify computer hardware components and assemble, disassemble and set up a working PC with the required operating system. | K3 |
| 2. | Apply basic network, browser, search, and cyber security skills to access the Internet, troubleshoot connectivity problems, and practice safe online communication. | K3 |
| 3. | Build formatted technical documents such as certificates, abstracts and newsletters using Word | K3 |
| 4. | Test for formulas, functions and charts on spreadsheets for data analysis, and design structured interactive presentations. | K4 |

**SYLLABUS**

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| 1 | **Identifying Components:**<br>Identify the peripherals of a computer and the components inside the CPU cabinet (motherboard, processor, RAM, SMPS, storage drives such as HDD/SSD/NVMe, and graphics card) and their functions. Draw the block diagram of the CPU with the configuration of each component. |
| 2 | **Assembling a PC** :<br>Disassemble a PC and reassemble it back to working condition. The lab instructor verifies the work and conducts a viva; refer to the demonstration video provided with the course content. |
| 3 | **Operating System Installation:**<br>Individually install Windows and Linux on a personal computer. The lab instructor verifies the installation and conducts a viva. |
| 4 | **Network Connectivity :**<br>Connect the system to a LAN/Wi-Fi and access the Internet by configuring TCP/IP settings (IP address, DNS). Demonstrate access to websites and email, and use basic troubleshooting commands (ping, ipconfig/ifconfig, tracert). |

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| 5 | **Browsers, Search and Netiquette:**<br>Customize a web browser with bookmarks, pop-up blockers and useful extensions (PDF viewer, ad blocker, password manager). Use search engines effectively with filters and Boolean operators, and demonstrate proper netiquette for online communication. |
| 6 | **Cyber Hygiene :**<br>Identify common online threats (phishing, malware, unsafe downloads) and configure the system for safe browsing: HTTPS-only browsing, strong passwords with a password manager, two-factor authentication, and keeping the OS and antivirus updated. |
| 7 | **Certificate Creation :**<br>Using MS Word , Create a project certificate covering fonts, drop cap, text effects, character spacing, borders and colours, headers and footers, and date/time options. |
| 8 | **Project Abstract:**<br>Create a project document on a topic of your choice, and demonstrate the use of formatting styles, tables, bullets and numbering, text direction, cell alignment, footnotes, hyperlinks, symbols, spell check, and track changes. |
| 9 | **Newsletter, and Introduction :**<br>Create a newsletter covering table of contents, newspaper columns, images and clip art, the drawing toolbar and WordArt, image formatting, text boxes, paragraph formatting and mail merge. |
| 10 | **Spreadsheet Scheduler and GPA:**<br>Using MS Excel, create a scheduler (gridlines, cell formatting, AutoFill, summation) and a GPA sheet using cell referencing, the AVERAGE, STDEV and COUNT functions, charts, and multiple worksheets. |
| 11 | **Spreadsheet Data Analysis:**<br>Perform data analysis using LOOKUP/VLOOKUP, splitting cells, freezing panes, grouping/outlining, sorting, Boolean and logical operators, conditional formatting and a basic pivot table. |
| 12 | **Interactive Presentation:**<br>Create an interactive presentation covering slide layouts, WordArt, bullets and autoshapes, hyperlinks, inserting images, audio, video, tables and charts; master layouts (slide, template, notes),views and backgrounds and design templates. |
| | |
| **Reference Books:** | |
| 1 | Computer Fundamentals, Anita Goel, Pearson India Education,2017 |
| 2 | P. K. Sinha and Priti Sinha, Computer Fundamentals, BPB Publications. |
| 3. | Rajaraman V., Fundamentals of Computers, PHI Learning. |
| 4. | Joan Lambert, Microsoft Office 2019/2021 Step by Step, Microsoft Press. |
| | |
| **Web References:** | |
| 1. | Virtual Labs – Computer Organisation and Hardware: https://www.vlab.co.in/broad-area-computerscience-and-engineering |
| 2. | AWS Free Tier: https://aws.amazon.com/free/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2610018 | MC | -- | -- | 2 | -- | 40 | -- | -- |

**NSS, SPORTS & YOGA**

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

**Course Objectives:**

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|---|---|
| 1. | The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service. |
| 2. | To make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality. |

**Course Outcomes**

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | Understand the importance of discipline, character and service motto. | K2 |
| 2. | Explore human relationships by analyzing social problems. | K3 |
| 3. | Develop leadership skills and civic responsibilities. | K3 |
| 4. | Understand the importance of yoga and sports for physical fitness and sound health | K2 |
| 5. | Demonstrate health related fitness components. | K2 |

**SYLLABUS**

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| **UNIT-I (10Hrs)** | **Orientation** : General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.<br>**Activities:**<br>i Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills<br>ii Conducting orientations programs for the students –future plans-activities-releasing road map etc.<br>iii Displaying success stories-motivational biopics- award winning movies on societal issues etc.<br>iv Conducting talent show in singing patriotic songs-paintings- any other contribution. |
| **UNIT-II (10 Hrs)** | **Nature & Care**<br>**Activities:**<br>i Best out of waste competition.<br>ii Poster and signs making competition to spread environmental awareness. |

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|---|---|
| | iii Recycling and environmental pollution article writing competition.<br>iv Organising Zero-waste day.<br>v Digital Environmental awareness activity via various social media platforms.<br>vi Virtual demonstration of different eco-friendly approaches for sustainable living.<br>vii Write a summary on any book related to environmental issues. |
| | |
| **UNIT-III (10 Hrs)** | **Community Service**<br>**Activities:**<br>i Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc.<br>i Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,<br>ii Conducting consumer Awareness. Explaining various legal provisions etc.<br>iii Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.<br>iv Any other programmes in collaboration with local charities, NGOs etc. |
| | |
| **UNIT-IV (10 Hrs)** | Yoga Activities:<br>Yoga practices- Asana, Surya Namaskar, Meditation. |
| | |
| **UNIT-V (10 Hrs)** | Physical Activities:<br>Practicing general and specific warm up, Aerobics. Cardio respiratory fitness, Skipping and Running. Practice one major game and one individual sport(volley ball, basket ball). |
| | |

| **Reference Books:** | |
|---|---|
| 1. | Nirmalya Kumar Sinha & Surajit Majumder, A Text Book of National Service Scheme Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6) |
| 2. | Red Book-National Cadet Corps-Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi |
| 3. | Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008 |
| 4. | Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007 |
| 5. | Ram Ahuja. Social Problems in India, Rawat Publications, New Delhi. |
| 6. | The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014 |
| **e-Resources** | |
| 1. | https://books.google.co.in/books/about/Social_Problems_in_India.html?id=jsvJoAEACAAJ&redi |

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| | r_esc=y |
| 2. | https://openlibrary.org/verify_human?next=/books/OL32142863M/Social_problems_in_India%3Futm_source%3Dchatgpt.com |



| **Regulation:R26** | **I/IV-B.Tech. II -Semester** |
|---|---|

**COMPUTER SCIENCE AND DESIGN**

**COURSE STRUCTURE**

**(With effect from 2026-27 admitted Batch onwards)**

| Course Code | Course Name | Category | L | T | P | Cr | C.I.E. | S.E.E. | Total Marks |
|---|---|---|---|---|---|---|---|---|---|
| A2620001 | Differential Equations and Vector Calculus | **BS** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2620003 | Applied Chemistry for Emerging Technologies | **BS** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2620004 | Data Structures using C | **ES** | 3 | 0 | 0 | 3 | 40 | 60 | 100 |
| A2620006 | Fundamentals of Python Programming | **ES** | 2 | 0 | 0 | 2 | 40 | 60 | 100 |
| A2620008 | Sustainable Development: Principles and practices | **ES** | 2 | 0 | 0 | 2 | 40 | 60 | 100 |
| A2620011 | Design Thinking and Innovation | **ES** | 0 | 1 | 3 | 2.5 | 50 | 50 | 100 |
| A2620013 | Data Structures using C Lab | **ES** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |
| A2620015 | Fundamentals of Python Programming Lab | **ES** | 0 | 0 | 2 | 1 | 50 | 50 | 100 |
| A2620018 | Applied Chemistry for Emerging Technologies Lab | **BS** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |
| A2620019 | English for Technical | **HS** | 0 | 0 | 3 | 1.5 | 50 | 50 | 100 |

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|---|---|---|---|---|---|---|---|---|---|
| | Communication and MOODLES Lab | | | | | | | | |
| A2620020 | Universal Human Values - II | **MC** | 2 | 0 | 0 | 0 | 50 | - | 50 |
| | | **TOTAL** | **15** | **1** | **14** | **21** | **500** | **550** | **1050** |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620001 | BS | 3 | -- | -- | 3 | 40 | 60 | 3 Hrs. |

**DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS**

(Common for all Branches)

**Course Objectives:** Students are expected to learn

| | |
|---|---|
| 1. | First order ordinary differential equations and some simple geometrical and physical applications. |
| 2. | Methods of solution of linear higher order ordinary differential equations. |
| 3. | Formation and solution of linear partial differential equations. |
| 4. | Concepts of Gradient, divergence, curl. |
| 5. | Vector integral theorems. |

**Course Outcomes:** After completion of the course, the student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | **Apply** the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits. | K3 |
| 2. | **Solve** linear ordinary differential equations of second and higher order which occur in various engineering fields. | K3 |
| 3. | **Apply** the appropriate methods of solution for partial differential equations that model physical processes. | K3 |
| 4. | **Compute** the gradient of a scalar field, curl and divergence of various vector fields to demonstrate their physical behavior. | K3 |
| 5. | **Calculate** the work done against a field, circulation and flux using vector calculus. | K3 |

**SYLLABUS**

| | |
|---|---|
| **UNIT-I (10Hrs)** | **Differential equations of first order and first degree**<br>Exact equations and equations reducible to exact form (Integrating Factor), Linear differential equations, Bernoulli's equations. Orthogonal Trajectories, Applications: Newton's Law of cooling, Law of natural growth and decay, Electrical circuits: RL & RC circuits. |
| **UNIT-II (10 Hrs)** | **Linear differential equations of higher order (Constant Coefficients)**<br>Definitions, homogenous and non-homogenous, complimentary function, particular integral, general solution, Method of variation of parameters – Wronskian. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion. |
| **UNIT-III (10 Hrs)** | **Partial Differential Equations**<br>Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous and Non-Homogeneous Linear Partial differential equations with |

| | |
|---|---|
| | constant coefficients. |
| | |
| **UNIT-IV (10 Hrs)** | **Vector Differentiation**<br>Scalar and vector point functions, vector operator Del, Gradient and applications, Directional derivative, Divergence – Solenoidal and Curl – Irrotational Field – Scalar Potential, Vectoridentities. |
| | |
| **UNIT-V (10 Hrs)** | **Vector Integration**<br>Line Integral – Circulation – Work done by force, surface integral – flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and their applications. |
| | |

| **Textbooks:** | |
|---|---|
| 1. | HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition. |
| 2. | AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition. |
| **Reference Books:** | |
| 1. | Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition. |
| 2. | Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018. |
| 3. | Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition. |
| 4. | Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint). |
| 5. | Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017 |
| 6. | Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty two Edition (Reprint 2022). |
| 7. | Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015. |
| | |
| **e-Resources** | |
| 1. | https://mathworld.wolfram.com/First-OrderOrdinaryDifferentialEquation.html |
| 2. | https://en.wikipedia.org/wiki/Differential_equation |
| 3. | https://en.wikipedia.org/wiki/Partial_differential_equation |
| 4. | https://en.wikipedia.org/wiki/Vector_calculus |
| 5. | https://en.wikipedia.org/wiki/Vector_calculus |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620003 | BS | 3 | -- | -- | 3 | 40 | 60 | 100 |

**APPLIED CHEMISTRY FOR EMERGING TECHNOLOGIES**

(Common to CSBS, CSD, CSIT)

**Course Objectives:**

| | |
|---|---|
| 1. | **Develop** an understanding of fundamental quantum chemistry concepts and molecular orbital theory relevant to advanced materials and electronic technologies. |
| 2. | **Evaluate** the electrochemical principles and operating characteristics of batteries, fuel cells . |
| 3. | **Analyze** the working principles and applications of electrochemical sensors, biosensors and spectroscopic techniques for chemical and material characterization. |
| 4. | **Explain** the chemistry, electronic properties and processing of semiconductor and superconducting materials used in electronic and emerging technologies |
| 5. | **Examine** the synthesis, properties and applications of nanomaterials in nanoelectronics, sensing and other emerging technologies with emphasis on sustainable materials development |

**Course Outcomes:** After successful completion of the course, the student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | **Apply** quantum chemistry principles and molecular orbital theory to analyze bonding, molecular structure and electronic characteristics of molecules relevant to advanced materials. | K3,K4 |
| 2. | **Evaluate** the electrochemical principles, performance characteristics and applications of batteries and fuel cells. | K3,K4 |
| 3. | **Analyze** the working principles and applications of electrochemical sensors, biosensors, UV-Visible and IR spectroscopy for chemical and material characterization. | K3,K4 |
| 4. | **Apply** principles of semiconductor chemistry, doping and fabrication processes to explain the properties and applications of semiconductor and superconducting materials in electronic technologies. | K3 |
| 5. | **Analyze** the synthesis methods, properties and applications of nanomaterials for nanoelectronics, sensing, memory and emerging electronic technologies**.** | K3,K4 |

**SYLLABUS**

| | |
|---|---|
| **UNIT-I (10Hrs)** | **Quantum Chemistry**<br>• Fundamentals of quantum chemistry: Black-body radiation, photoelectric effect, wave-particle duality, de Broglie wave equation, Heisenberg's uncertainty principle.<br>• Fundamentals of quantum mechanics: Schrödinger wave equation (time independent), significance of $\Psi$ and $\Psi^2$; particle in a one-dimensional box.<br>• Molecular orbital theory: bonding in homo-and hetero-nuclear di atomic molecules; energy-level diagrams of $O_2$, $N_2$, HF and CO ; calculation of bond order<br>• π-molecular orbitals of butadiene and benzene. |

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| **UNIT-II (10 Hrs)** | **Triad of Electrochemical Power**<br>• **Fundamentals of electrochemistry**: Galvanic and electrolytic cells, Electrode potential, Nernst equation, Types of electrodes-ion-selective and glass electrode, Reference electrodes -Calomel electrode, (Construction and working), Electro-chemical series and its uses. Conductometric titrations: strong acid vs strong base titrations.<br>• Batteries: Introduction, classification of batteries, characteristics of battery-capacity, power density, cell balancing and cycle life.<br>• Primary batteries: construction, working and applications of zinc-air and Lithium batteries<br>• Secondary batteries: lead-acid, Lithium-ion batteries—construction, charging-discharging reactions and applications.<br>• Fuel cells: Hydrogen-oxygen fuel cells (Polymer electrolyte fuel cells (PEM), solid-oxide fuel cell(SOFCs)) working, applications and limitations |
| | |
| **UNIT-III (10 Hrs)** | **Sensors, Biosensors and Principles of Spectroscopic Techniques**<br>• Sensors: Introduction, terminologies- Transducer, Actuators and Sensors, working principle and applications of electrochemical sensor,(Ampherometric sensors, Potentiometric sensors) .<br>• Biosensors: Components of a Biosensor and applications.<br>• Colorimetric Method: Electromagnetic spectrum, Beer-Lambert's law, Principle and instrumentation of colorimeter, application of colorimeter in iron estimation.<br>• Spectroscopy: UV-Visible spectroscopy: principle, electronic transitions, instrumentation, applications.<br>• Infrared (IR) spectroscopy: Introduction, principle, mode of vibrations, instrumentation and applications. |
| | |
| **UNIT-IV (10 Hrs)** | **Chemistry of Semiconductors and Superconductors**<br>• Intrinsic and extrinsic semiconductors, Preparation of pure semiconductor material (Zone refining), Chemistry of compound semiconductors (GaAs, InP, SiC), Advanced semiconducting materials-Organic(Introduction, applications,Conduction mechanism of Polyacetylene) and Perovskite semiconductors (Definition,applications)<br>• Solar photovoltaic cell (PV cell): construction, working principle and applications.<br>• Photolithography: Basic principle, applications<br>• Superconductors: Characteristics, Type-I and Type-II, Low temperature and high temperature superconductors, superconducting materials (NbTi, $Nb_3Sn$), applications |
| | |
| **UNIT-V** | **Chemistry of nanomaterials** |

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|---|---|
| **(10 Hrs)** | • Introduction to Nano chemistry, carbon nanotubes (CNTs), graphene; synthesis methods [chemical vapor deposition (CVD, Arc -discharge method), Applications.<br>• Quantum dots: Synthesis (Sol-Gel), properties, and real time applications (biological imaging and sensing).<br>• Metallic nanoparticles: Preparation (Biological process) and applications of gold, silver nanoparticles.<br>• Applications of nanomaterials in nanoelectronics, interconnects, sensors (optical and biosensors), memory devices.<br>• Materials of Spintronics: Magnetic nanoparticles, ferromagnetic materials, topological insulators.(Basic Concept and applications) |

| | |
|---|---|
| **Textbooks:** | |
| 1. | Jain, P. C., & Jain, M. (2020). Engineering Chemistry (17th ed.). Dhanpat Rai Publishing Company |
| 2. | Agarwal, S. (2026). Engineering chemistry (5th ed.). Cambridge University Press |
| **Reference Books:** | |
| 1. | Linden, D., & Reddy, T. B. (Eds.). (2001). Handbook of batteries (3rd ed.). McGraw-Hill. |
| 2. | Rao, C. N. R., Müller, A., & Cheetham, A. K. (Eds.). (2007). The chemistry of nanomaterials: Synthesis, properties and applications (Vols. 1 2). Wiley-VCH. |
| 3. | Sharma, B. K., & Gaur, H. (2019). Engineering chemistry. Krishna Prakashan Media |
| 4. | Streetman, B. G., & Banerjee, S. K. (2015). Solid state electronic devices (7th ed.). Pearson Education. |
| **e-Resources** | |
| 1. | nptel.ac.in/courses/115105131 |
| 2. | https://archive.nptel.ac.in/content/storage2/courses/103105065/M11l11.pdf |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620004 | ES | 3 | -- | -- | 3 | 40 | 60 | 3 Hrs |

**DATA STRUCTURES USING C**

(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, IT)

**Course Objectives:**

| | |
|---|---|
| 1. | Explain the fundamental concepts of data structures, abstract data types and basic algorithm complexity. |
| 2. | Demonstrate searching and sorting techniques for efficient data processing. |
| 3. | Implement linked lists, stacks and queues using appropriate operations. |
| 4. | Demonstrate the use of tree structures for data organization and retrieval. |
| 5. | Use graph and hashing techniques for data representation and processing. |

**Course Outcomes:** At the end of the course, student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | **Apply** appropriate searching and sorting techniques to solve data management problems. | K3 |
| 2. | **Apply** linked list operations to represent and process dynamically organized data. | K3 |
| 3. | **Apply** stacks and queues to solve problems involving data processing and expression evaluation**.** | K3 |
| 4. | **Implement** tree-based algorithms for efficient searching and data organization. | K3 |
| 5. | **Apply** graph and hashing algorithms for network connectivity problems and efficient data retrieval. | K3 |

**SYLLABUS**

| | |
|---|---|
| **UNIT-I (10Hrs)** | **Introduction to data Structures**: Abstract Data Types (ADT), Need and classification of Data Structures, Introduction to algorithm analysis: Time and space complexity (basic concepts).<br>**Searching and Sorting:** Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort. |
| **UNIT-II (10 Hrs)** | **Linear Data Structures:** Dynamic memory allocation, Recursion, Linked Lists: Singly, Doubly, and Circular Linked Lists, Basic operations (creation, insertion, deletion, traversal), Applications of Linked Lists. |
| **UNIT-III (10 Hrs)** | **Stacks:** Array and Linked List implementations, operations on stacks, Basic applications: infix to postfix conversion, postfix expression evaluation, parenthesis matching.<br>**Queues:** Simple Queue, Circular Queue, operations and Applications of Queues. |
| **UNIT-IV (10 Hrs)** | **Trees:** Introduction and terminology, Binary Trees, Tree representation, Tree traversals, **Binary Search Trees**: Insertion, Deletion, Searching, **Heap**: Min Heap and Max Heap, |

| | |
|---|---|
| | Applications of Trees. |
| | |
| **UNIT-V (10 Hrs)** | **Graphs**: Graph terminology, Graph representations: Adjacency Matrix and Adjacency List, Graph traversals: Breadth First Search (BFS) and Depth First Search (DFS), Introduction to Minimum Spanning Trees (Prim's Algorithm), Applications of Graphs.<br>**Hashing**: Introduction to Hashing, Hash Functions, Collision Resolution, Open Addressing: linear probing, quadratic probing, double hashing, separate chaining, rehashing, examples |
| | |

| **Textbooks:** | |
|---|---|
| 1. | Balagurusamy, E. (2025). Data Structures Using C (2nd ed.). McGraw Hill Education India. |
| 2. | Thareja, R. (2023). Data Structures Using C (3rd ed.). Oxford University Press India. |
| **Reference Books:** | |
| 1. | Deshpande, P. S., & Kakde, O. G. (2003). C and data structures. Dreamtech Press. |
| 2. | Shukla, R. K. (2009). Data structures using C and C++. Wiley India. |
| | |
| **e-Resources** | |
| 1. | National Programme on Technology Enhanced Learning. (2026). Programming, Data Structures and Algorithms using C. IIT Madras. https://nptel.ac.in/courses/106106127 |
| 2. | Virtual Labs: https://ds1-iiith.vlabs.ac.in/ |
| 3. | Programiz. Data Structures. https://www.programiz.com/dsa |
| 4. | Visualgo. Data Structure and Algorithm Visualizations. https://visualgo.net/en |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620006 | ES | 2 | -- | -- | 2 | 40 | 60 | 100 |

**FUNDAMENTALS OF PYTHON PROGRAMMING**

**(Common to CSBS, CSD, CSIT)**

**Course Objectives:**

| | |
|---|---|
| 1. | Develop proficiency in Python control structures while explicitly linking conditionals to abstraction and loops to pattern recognition. |
| 2. | Build competence in selecting Python data structures (strings, lists, tuples, dictionaries) |
| 3. | Introduce object-oriented programming concepts such as classes, objects, inheritance, and polymorphism, enabling students to model real-world problems through encapsulation and abstraction. |
| 4. | Train students to decompose problems into functions and recursive sub-problems, use lambda expressions and functional tools, reason informally about algorithmic efficiency, and use AI coding assistants responsibly. |
| 5. | Equip students with file handling, exception handling, and basic data analysis/visualization skills, framed as decomposing a larger data workflow into manageable, robust steps and to integrate these skills in a capstone mini-project. |

**Course Outcomes:** At the end of the course student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | Develop Python programs using variables, data types, operators, input/output, conditional statements, loops, and appropriate coding practices to solve computational problems. | K3 |
| 2. | Implement Python programs using strings, lists, tuples, dictionaries, and comprehensions to efficiently represent, manipulate, and process different types of data. | K3 |
| 3. | Develop and analyze modular Python programs using functions, lambda expressions, functional programming tools, recursion, modules, and libraries, considering basic algorithmic efficiency. | K4 |
| 4. | Design and implement object-oriented Python programs using classes, objects, encapsulation, inheritance, polymorphism, operator overloading, and composition to model real-world problems. | K4 |
| 5. | Develop Python applications that perform file processing and exception handling, and apply NumPy, pandas, and Matplotlib for basic data analysis and visualization. | K3 |

**SYLLABUS**

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|---|---|
| **UNIT-I (10Hrs)** | **PYTHON BASICS AND CONTROL FLOW**<br>**Python Basics:** Translating algorithms and flowcharts into Python, variables, data types, operators and input/output. |

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|---|---|
| | Good Coding Practice: Docstrings, inline comments and print-based debugging (threaded through every unit from here on).<br>**Conditional Statements:** if, if-else and chained/nested conditionals, abstraction over multiple cases.<br>**Loops:** for and while loops, pattern recognition of repetition.<br>**Loop Control:** Nested loops and loop control (break, continue, pass), decomposition of complex repetitive logic. |
| | |
| **UNIT-II (10 Hrs)** | **DATA STRUCTURES IN PYTHON**<br>**Strings**: Pattern matching and abstraction of text as a sequence; string operations and methods.<br>**Lists:** Abstraction of ordered mutable collections; list operations; list comprehension.<br>**Tuples and Dictionaries:** Abstraction of fixed (tuples) and key-based (dictionaries) data; common operations and use-case selection. |
| | |
| **UNIT-III (10 Hrs)** | **FUNCTIONS, RECURSION AND MODULES**<br>**Functions:** Decomposition of a program into independent, reusable sub-problems; parameters, return values and scope.<br>**Lambda Expressions and Functional Tools:** lambda, map, filter, and reduce, concise transformations and their use cases.<br>**Recursion:** Algorithmic thinking applied to self-similar sub-problems.<br>Algorithmic Efficiency: Informal comparison of iterative vs. recursive approaches and their time trade-offs.<br>**Modules and Libraries:** Importing and creating modules, packages and built-in libraries. |
| | |
| **UNIT-IV (10 Hrs)** | **OBJECT ORIENTED PROGRAMMING**<br>**Introduction to OOP**: Features of object-oriented programming, encapsulation, abstraction, inheritance, polymorphism.<br>**Classes and Objects:** Class definition, the __init__ method, class variables and instance variables, public and private data members, calling methods from other methods, built-in class attributes.<br>**Inheritance:** Inheriting classes, types of inheritance, method overriding.<br>**Polymorphism and Operator Overloading:** Polymorphism through method overriding; operator overloading. |
| | |
| **UNIT-V (10 Hrs)** | **FILE HANDLING, EXCEPTION HANDLING**<br>**File Handling:** Reading and writing files; decomposition of a data workflow into read/process/write stages.<br>**Exception Handling**: try-except-finally; abstraction of error cases into predictable, manageable patterns; user-defined exceptions. |
| | |
| **Textbooks:** | |

| | |
|---|---|
| 1. | Downey, A. B. (2015). Think Python: How to think like a computer scientist (2nd ed.). O'Reilly Media. |
| 2. | J. V. Guttag, Introduction to Computation and Programming Using Python: With Application to Computational Modeling and Understanding Data, 3rd Edition, MIT Press, 2021. |
| **Reference Books:** | |
| 1. | Sweigart, A. (2025). Automate the boring stuff with Python: Practical programming for total beginners (3rd ed.). No Starch Press. |
| 2. | McKinney, W. (2022). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media. |
| 3. | J. M. Wing, Computational Thinking, Communications of the ACM, 49(3), 33–35, 2006. https://doi.org/10.1145/1118178.1118215 |
| 4. | Jeeva Jose, "Taming Python by Programming", 2nd Edition, Khanna Book Publishing Co., Delhi, 2019. |
| | |
| **e-Resources** | |
| 1. | Python Software Foundation. (n.d.). The Python tutorial. https://docs.python.org/3/tutorial/ |
| 2. | W3Schools. (n.d.). Python tutorial. https://www.w3schools.com/python/ |
| 3. | NPTEL. (n.d.). Programming, data structures and algorithms using Python. https://nptel.ac.in |
| 4. | Real Python. (n.d.). Python tutorials. https://realpython.com/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| **A2620011** | **ES** | **--** | **1** | **3** | **2.5** | **50** | **50** | **3 Hrs.** |

**Design Thinking and Innovation**

**(Common to AIDS, CSD, CSIT, IT)**

**Course Objectives:** At the end of the course, the teacher should be able to:

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|---|---|
| 1 | Introduce students to the principles, mindset and methodologies of **Design Thinking and human-centered problem solving**. |
| 2 | Develop students' ability to **observe, empathize, analyze user needs and formulate well-defined engineering and societal problems**. |
| 3 | Develop students' **creative and innovative thinking abilities** through systematic ideation, concept generation and evaluation techniques. |
| 4 | Enable students to **develop, prototype, test and iteratively improve solutions** using user feedback and appropriate physical/digital tools. |
| 5 | Develop students' ability to identify innovation opportunities, create value propositions, understand basic IPR and communicate innovative solutions effectively. |

**Course Outcomes**: *After successful completion of the course, student will be able to*

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1 | Explain the principles, processes and mindset of Design Thinking for engineering problem-solving and innovation | K2 |
| 2 | Apply empathy, observation and problem-framing techniques to identify user needs and define engineering problems | K3 |
| 3 | Apply creative and systematic ideation techniques to generate and select innovative solution concepts | K3 |
| 4 | Analyze solution concepts and prototypes using user feedback and appropriate design criteria for iterative improvement | K4 |
| 5 | Analyze the innovation potential of solutions considering value proposition, feasibility, sustainability, societal impact and basic IPR aspects | K4 |

**SYLLABUS**

| | |
|---|---|
| 1 | **UNIT I — Design Thinking Foundations & Empathy**<br>Introduction to Design Thinking ; Design thinking mindset and principles ; Design thinking vs. conventional engineering problem-solving ; Human-centered design ; Five-stage design thinking process ; Creativity, curiosity and observation ; Understanding users and stakeholders ; Empathy and user research ; Interviews, observation and contextual inquiry ; Empathy mapping ; Personas and user journeys ; Identifying explicit and latent/unarticulated needs ; Ethical considerations in user research<br>**Practical Activities** |

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|---|---|
| | • Observation of a real-life problem<br>• Conducting user interviews<br>• Preparing an empathy map<br>• Developing a persona<br>• Preparing a user journey map<br>**Unit Outcome**<br>Students will be able to **understand users, identify their needs and translate observations into meaningful design opportunities** |
| 2 | **UNIT II — Problem Definition, Analysis & Opportunity Identification**<br>From observations to insights ; Data synthesis and affinity mapping ; Identifying patterns and pain points ; Problem statements ; Point-of-View (POV) statements ; "How Might We" (HMW) questions ; Stakeholder mapping ; 5 Whys and root-cause analysis ; Problem prioritization ; Need–Opportunity mapping ; Problem validation ; Design constraints and specifications ; Social, environmental and sustainability considerations ; Identifying opportunities for innovation<br>**Practical Activities**<br>• Affinity mapping of interview data<br>• Develop three alternative problem statements<br>• Prepare POV and HMW statements<br>• Stakeholder map<br>• Root-cause analysis<br>• Select and validate one problem/opportunity<br>**Unit Outcome**<br>Students will be able to **analyze user research, identify root causes, frame well-defined problems and recognize opportunities for innovative solutions**. |
| 3 | **UNIT III — Creativity, Ideation & Concept Development**<br>Creativity and innovation mindset; Divergent and convergent thinking ; Brainstorming and brainwriting ; SCAMPER ; Mind mapping ; Reverse brainstorming ; Analogical thinking ; Biomimicry<br>; Lateral thinking ; Forced connections ; Morphological analysis ; TRIZ — basic principles ; Individual and collaborative ideation ; AI-assisted ideation: opportunities and limitations ; Idea clustering and screening ; Impact–feasibility matrix ; Concept selection ; Design criteria and decision matrix<br>**Practical Activities**<br>• Generate 20–30 ideas for a selected problem<br>• SCAMPER exercise<br>• Brainwriting exercise<br>• Biomimicry-based ideation<br>• AI-assisted idea generation and critical evaluation<br>• Prepare an idea-selection matrix<br>• Select the top 2–3 concepts<br>**Unit Outcome**<br>Students will be able to **generate diverse solution concepts using creative techniques and** |

| | |
|---|---|
| | **systematically select promising ideas based on defined criteria**. |
| 4 | **UNIT IV — Prototyping, Experimentation & Validation**<br>Concept development ; Prototype mindset ; Types of prototypes: Paper prototype, Physical prototype, Digital prototype, Storyboard, Role-play, Service prototype ; Low-fidelity and high-fidelity prototypes<br>; Rapid prototyping ; "Fail fast, learn fast" ; Minimum Viable Prototype (MVP) ; Design experimentation ; Prototype evaluation ; User testing ; Think-aloud method ; Feedback collection ; Usability and functionality testing; Iterative design ; Design critique ; Documentation of iterations ; Use of CAD, simulation and AI-assisted visualization tools<br>**Practical Activities**<br>• Develop a low-fidelity prototype<br>• Prepare a storyboard<br>• Build a physical/digital prototype<br>• Conduct user testing<br>• Collect structured feedback<br>• Modify the prototype<br>• Prepare an iteration log |
| 5 | **UNIT V — Innovation, Value Creation & Communication**<br>**Innovation: meaning, characteristics and importance:** Invention vs. innovation ; Incremental, disruptive and radical innovation ; Product, process, service and social innovation ; Technology-driven vs. need-driven innovation ; Sustainable and frugal innovation ; Innovation opportunities and emerging technologies ; From prototype to innovation ; Desirability, feasibility and viability ; Value Proposition<br>; Value Proposition Canvas ; Introduction to Business Model Canvas ; Customer segments and stakeholder value; Innovation validation; Market/user validation ; Intellectual Property Rights — introduction; Patents, copyright, trademarks and design protection ; Responsible innovation ; Inclusive and sustainable innovation ; Innovation ecosystem and entrepreneurship ; Communicating innovative ideas ; Innovation pitch<br>**Practical Activities**<br>1. Classify existing products/services as incremental, disruptive or radical innovations.<br>2. Identify three innovation opportunities from a campus/community problem.<br>3. Prepare a **Value Proposition Canvas**.<br>4. Validate the proposed innovation with potential users.<br>5. Prepare an introductory **Business Model Canvas**.<br>6. Identify possible IPR protection.<br>Prepare and deliver a **3-minute Innovation Pitch** |
| | |
| **Text Books**: | |
| 1 | Chandan Adhikari, Prabir Kumar Das & Tanay Pramanik, Textbook of Design Thinking and Innovation, S. Chand Publishing, 2026. |
| 2 | Shalini Rahul Tiwari & Rohit Rajendra Swarup, Design Thinking: A Comprehensive Textbook, Wiley India, 2024 |

| **Reference Books:** | |
|---|---|
| 1 | Soni, Pavan, Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving, Penguin Random House India. |
| 2 | Jauhari, Vinnie & Sudhanshu Bhushan, Innovation Management, Oxford University Press India. |
| **Web References**: | |
| 1 | Atal Innovation Mission, NITI Aayog – Innovation and Design Thinking Resources. https://aim.gov.in/ |
| 2 | Institution's Innovation Council (IIC), Ministry of Education, Government of India, https://iic.mic.gov.in/ |
| 3 | Startup India, Government of India – Innovation and Entrepreneurship Ecosystem, https://www.startupindia.gov.in/ |
| 4 | National Innovation Foundation – India – Grassroots and Frugal Innovation, https://nif.org.in/ |
| 5 | Stanford d.school – Design Thinking Bootleg and Design Thinking Resources, https://dschool.stanford.edu/resources/design-thinking-bootleg |
| 6 | IDEO.org – Human-Centered Design and Social Innovation Resources, https://www.ideo.org/ |
| 7 | IDEO U – Design Thinking and Innovation Resources, https://www.ideou.com/pages/design-thinking |
| 8 | NPTEL – Online Courses on Design, Innovation, Creativity and Entrepreneurship, https://nptel.ac.in/ |
| 9 | SWAYAM – Online Courses on Innovation, Entrepreneurship and Design Thinking, https://swayam.gov.in/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620013 | ES | -- | -- | 3 | 1.5 | 50 | 50 | 3 Hrs |

**DATA STRUCTURES USING C LAB**

(Common to AIDS, AIML, CIC, CSD, CSE, CSIT, IT)

**Course Objectives:** The main objectives of the course are to

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|---|---|
| 1 | Develop the ability to implement fundamental data structures and their operations using C programming. |
| 2 | Strengthen the ability to trace, debug and test code, and to choose an appropriate data structure and implementation strategy for a given problem. |

**Course Outcomes:** At the end of the course, student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1 | **Implement** and analyze array-based searching and sorting techniques. | K4 |
| 2 | **Apply** Linear Data Structures for data representation and manipulation. | K3 |
| 3 | **Apply** tree data structures to implement and solve non-linear data processing problems. | K3 |
| 4 | **Implement** hashing techniques for efficient data storage and retrieval operations. | K3 |

**SYLLABUS**

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|---|---|
| 1 | **Exercise 1: Array-Based Techniques, searching**<br>1. Write a program using pointers to reverse an array in place, without using a second array.<br>2. Write a program to dynamically allocate an array using malloc, resize it using realloc, and free it correctly.<br>3. Implement linear search on an unsorted array; count and print the number of comparisons made.<br>4. Implement binary search on a sorted array; count the comparisons made and compare the count with linear search on the same data. |
| 2 | **Exercise 2: Sorting**<br>1. Implement bubble, selection and insertion sort; print the array after each pass, and tabulate the comparisons and swaps each algorithm performs on the same input.<br>2. Implement Quick sort and Merge sort; print the array after every partition/merge step to observe how each algorithm progresses. |
| 3 | **Exercise 3: Singly Linked List**<br>1. Implement a singly linked list with creation, traversal, insertion and deletion of nodes.<br>2. Implement a singly linked list with insertion at the beginning, end and a given position.<br>3. Implement deletion of a node by value and by position with proper handling of an empty list, then reverse the list iteratively and recursively. |

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| 4 | **Exercise 4: Singly Linked List Applications**<br>1. Detect and remove duplicate nodes from an unsorted linked list.<br>2. Find the middle node of a singly linked list in a single traversal.<br>3. Merge two sorted singly linked lists into a single sorted list. |
| 5 | **Exercise 5: Doubly and Circular Linked Lists**<br>1. Implement a doubly linked list with insertion, deletion and forward/backward traversal.<br>2. Implement a circular linked list and demonstrate insertion, deletion and traversal without infinite looping.<br>3. Represent two polynomials using linked lists and write a program to add them. |
| 6 | **Exercise 6: Stack Implementation**<br>1. Implement a stack using an array and using a linked list, with overflow/underflow checks.<br>2. Write a program to check whether an expression has balanced parentheses using a stack. |
| 7 | **Exercise 7: Stack Applications**<br>1. Write a program to convert an infix expression to postfix using a stack.<br>2. Write a program to evaluate a postfix expression using a stack.<br>3. Write a program to check whether a given string is a palindrome using a stack. |
| 8 | **Exercise 8: Recursion and Stack Simulation**<br>1. Write recursive programs for factorial, Fibonacci and sum of digits.<br>2. Simulate the run-time call stack of a recursive function by writing an equivalent iterative version using an explicit stack. |
| 9 | **Exercise 9: Queue and Circular Queue**<br>1. Implement a circular queue and demonstrate that it avoids the false-overflow problem of a linear queue.<br>2. Implement a queue using an array and using a linked list. |
| 10 | **Exercise 10: Binary Trees**<br>1. Implement a Binary Tree with insertion and search operations.<br>2. Implement deletion of a node from a binary tree, handling all three cases (leaf, one child, two children). |
| 11 | **Exercise 11: Binary Search Trees**<br>1. Implement a Binary Search Tree (BST) with insertion and search operations.<br>2. Implement deletion of a node from a BST, handling all three cases (leaf, one child, two children). |
| 12 | **Exercise 12: Tree Traversals and Problems**<br>1. Implement a Binary Search Tree (BST) with insertion, and perform in-order, pre-order and post-order traversals.<br>2. Write a program to find the height of a binary tree and count the total number of nodes and leaf nodes.<br>3. Write a program to check whether a given binary tree is a valid Binary Search Tree. |

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| 13 | **Exercise 13: Hashing**<br>1. Implement a hash table using chaining for collision resolution, with insertion, search and deletion.<br>2. Implement a hash table using linear probing. |
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| **Web References:** | |
| 1 | NPTEL – Programming, Data Structures and Algorithms using C (IIT Madras) https://www.nptel.ac.in/courses/106106127 |
| 2 | NPTEL – Data Structures and Algorithms (IIT Delhi) https://www.nptel.ac.in/courses/106102064 |
| 3 | NPTEL – Programming, Data Structures and Algorithms using C – Course Archive https://onlinecourses-archive.nptel.ac.in/noc18_cs25/preview |
| 4 | Virtual Labs – Data Structures I (IIIT Hyderabad) https://ds1-iiith.vlabs.ac.in/ |
| 5 | Virtual Labs – IIIT Hyderabad Data Structures Labs https://www.vlab.co.in/participating-institute-iiit-hyderabad |
| 6 | GeeksforGeeks – Data Structures and Algorithms https://www.geeksforgeeks.org/dsa/ |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620015 | ES | -- | -- | 2 | 1 | 50 | 50 | 3 Hrs |

**FUNDAMENTALS OF PYTHON PROGRAMMING LAB**

**(Common to CSBS, CSD, CSIT)**

**Course Objectives:**

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|---|---|
| 1 | Develop proficiency in Python control structures while explicitly linking conditionals to abstraction and loops to pattern recognition. |
| 2 | Build competence in selecting Python data structures (strings, lists, tuples, dictionaries) |
| 3 | Introduce object-oriented programming concepts such as classes, objects, inheritance, and polymorphism, enabling students to model real-world problems through encapsulation and abstraction. |
| 4 | Train students to decompose problems into functions and recursive sub-problems, use lambda expressions and functional tools, reason informally about algorithmic efficiency, and use AI coding assistants responsibly. |
| 5 | Equip students with file handling, exception handling, and basic data analysis/visualization skills, framed as decomposing a larger data workflow into manageable, robust steps and integrate these skills in a capstone mini project. |

**Course Outcomes:**

| S. No | Outcome | Knowledge Level |
|---|---|---|
| 1 | Develop Python programs using variables, data types, operators, input/output, conditional statements, loops, and appropriate coding practices to solve computational problems. | K3 |
| 2 | Implement Python programs using strings, lists, tuples, dictionaries, and comprehension to efficiently represent, manipulate, and process different types of data. | K3 |
| 3 | Develop and analyze modular Python programs using functions, lambda expressions, functional programming tools, recursion, modules, and libraries, considering basic algorithmic efficiency. | K4 |
| 4 | Design and implement object-oriented Python programs using classes, objects, encapsulation, inheritance, polymorphism, operator overloading, and composition to model real-world problems. | K4 |
| 5 | Develop Python applications that perform file processing and exception handling, and apply NumPy, pandas, and Matplotlib for basic data analysis and visualization. | K3 |

**SYLLABUS**

**Exercise 1: Getting Started with Python**

1. Write and run a Python program that prints "Hello, World!" along with the Python version in use. Note the difference between running a saved script from the terminal and typing the same

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| | statements in the interactive shell.<br>2. Write a Python program that reads a name and an age from the user and prints a personalized greeting. |
| | 3. Write a Python program that reads two numbers and prints their sum, difference, product, quotient and remainder, each on a labelled line.<br>4. Write a Python program to compute simple interest given principal, rate, and time.<br>5. Write a Python program to compute the area and perimeter of a rectangle from user input.<br>6. Write a Python program to swap two numbers without using a third variable. |
| 2 | **Exercise 2: Conditionals**<br>1. Write a Python program to compute the Body Mass Index (BMI) from height and weight, rounded to two decimal places, and classify it as Underweight, Normal, Overweight or Obese using if-elif-else.<br>2. Write a menu-driven Python program to convert the temperature between Celsius, Fahrenheit and Kelvin based on the user's choice.<br>3. Write a Python program to find the largest of three numbers.<br>4. Write a Python program to Check whether a given year is a leap year. |
| 3 | **Exercise 3: Write a Python program to**<br>1. Determine whether a number is positive, negative, or zero.<br>2. Check if a given number is prime<br>3. Convert temperature between Celsius and Fahrenheit based on user choice<br>4. Compute and display the area of a valid triangle using Heron's formula, with inline comments explaining each step. |
| 4 | **Exercise 4: Loops, Number Problems and Patterns**<br>1. Use a for loop to print the first n Fibonacci terms, and a while loop to test whether a number is prime (using break, printing the number of division tests).<br>2. Use a while loop to compute the GCD and LCM of two integers using Euclid's algorithm.<br>3. Find the first n prime numbers using nested loops with break.<br>4. Sum of first N natural numbers |
| 5 | **Exercise 5: String Processing**<br>1. Write a function that checks whether a string is a palindrome using slicing and string methods, ignoring spaces, punctuation and case; justify whether a string or a list of characters is the better structure.<br>2. Write a script that counts the vowels, consonants, digits and special characters in a user-provided paragraph.<br>3. Write a script that sanitizes a full name with irregular capitalization and excess spaces using. strip () and. title () and extracts the initials. |
| 6 | **Exercise 6: Lists and List Comprehensions**<br>1. Remove duplicates from a list of sensor readings, first without set() and then with set(); compare the two approaches for time complexity and readability.<br>2. Use a list comprehension to filter out noise readings outside a realistic range (e.g., -40°C to 85°C). |

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| | 3. Compute and display the average, maximum and minimum of the filtered list. |
| 7 | **Exercise 7: Write a Python Program to**<br>1. Find the second-largest element in a list of exam scores.<br>2. Merge two sorted lists of student roll numbers into a single sorted list.<br>3. Generate squares of all even numbers from 1 to 50 using list comprehension. |
| 8 | **Exercise 8: Dictionaries and Tuples**<br>1. Create a dictionary of student records keyed by roll number with (Name, Marks) tuples as values; justify why a dictionary is preferable to a list of tuples for lookup.<br>2. Build a word-frequency dictionary from a text paragraph (case-insensitive). (Count the frequency of each word in a paragraph using a dictionary)<br>3. Search the student dictionary by roll number, assign grades and identify the highest scorer. |
| 9 | **Exercise 9: Recursion vs Iteration**<br>1. Write a recursive function Fibonacci recursive(n) that returns the n-th Fibonacci term.<br>2. Write an iterative function Fibonacci iterative(n) using a loop.<br>3. Compute the GCD of two numbers recursively using the Euclidean algorithm.<br>4. Compute factorial using recursion |
| 10 | **Exercise 10: File Handling and Modular Workflow**<br>1. Create a file student_records.txt and write five lines of data formatted as Roll No, Name, Marks.<br>2. Count the words, lines, and characters in a given text file.<br>3. Read a text file of daily temperature readings and write the maximum, minimum, and average to an output file |
| 11 | **Exercise 13: Exception Handling**<br>1. Write a program to compute the real roots of a quadratic equation, handling all cases (real distinct, real equal, no real roots).<br>2. Wrap the parsing and calculation in a try-except block handling Value Error and Zero Division Error, with a finally block logging attempts to calculation_log.txt.<br>3. Package the logic into a reusable function solve quadratic (a, b, c) suitable for import by other modules. |
| | |
| **Reference Books:** | |
| 1 | Sweigart, A. (2025). Automate the boring stuff with Python: Practical programming for total beginners (3rd ed.). No Starch Press. |
| 2 | McKinney, W. (2022). Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media. |
| 3 | Wing, J. M. (2006). Computational thinking. Communications of the ACM, 49(3), 33–35. https://doi.org/10.1145/1118178.1118215 |
| **Web References:** | |
| 1 | Python Software Foundation – The Python Tutorial: https://docs.python.org/3/tutorial/ |
| 2 | NPTEL – Python for Data Science: https://nptel.ac.in |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620018 | BS | -- | -- | 3 | 1.5 | 50 | 50 | 100 |

**APPLIED CHEMISTRY FOR EMERGING TECHNOLOGIES LAB**

(Common to CSBS, CSD, CSIT)

**Course Objectives:**

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| 1 | To develop practical skills in chemical analysis using volumetric, electrochemical, spectroscopic, and instrumental techniques for the characterization of chemical, water, and electronic-grade samples. |
| 2 | To develop hands-on skills in emerging chemistry applications through the synthesis and characterization of nanomaterials, polymeric dispersions, and electrochemical biosensors, and to relate experimental observations to technological applications. |

**Course Outcomes:** After successful completion of the course, the student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1 | **Apply** volumetric, conductometric, potentiometric, and pH-metric techniques to determine the concentration and properties of chemical and battery electrolyte samples. | K3 |
| 2 | **Apply** analytical and instrumental techniques to determine water quality parameters and metal-ion concentrations in water and electronic-grade samples. | K3 |
| 3 | **Apply** spectroscopic, electrochemical, and chemical methods to demonstrate Beer–Lambert law, identify functional groups, construct biosensors, and characterize chemical species. | K3 |
| 4 | **Apply** chemical and computational/virtual-laboratory techniques to synthesize nanomaterials, evaluate polymer/nanomaterial dispersions, and determine electronic energy levels using the particle-in-a-box model. | K3 |

**SYLLABUS**

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|---|---|
| 1 | Determination of the concentration/strength of sulfuric acid in Pb-acid battery electrolyte |
| 2 | Conductometric titration of a strong acid against a strong base. |
| 3 | Determination of a strong acid against a strong base by using pH meter. |
| 4 | Assemble a simple Zn-carbon (or lead-acid) cell and measure its voltage and approximate capacity |
| 5 | Determination of weak acid against a strong base by using potentiometer |
| 6 | Estimation of Iron in Mohr's salt by potassium dichromate |
| 7 | Determination of the specific and equivalent conductance of an electrolyte solution. |
| 8 | Determination of the pH of an electronic-grade (deionized) water sample used in wafer/PCB cleaning |
| 9 | Construction and performance evaluation of a simple enzyme based electrochemical biosensor for glucose detection (demo). |

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| 10 | Determination of the concentration of $Fe^{2+}/Fe^{3+}$ in a given water or leach liquor sample by colorimetry |
| 11 | Verification of Beer-Lambert's Law by colorimetry. |
| 12 | Synthesis of nanomaterial by chemical precipitation method |
| 13 | Identify functional groups present in a given organic or mineral-bound sample from its IR spectrum. (virtual lab) |
| 14 | Determination of the viscosity of a polymer/nanomaterial-dispersion solution using an Ostwald/ Redwood viscometer. |
| 15 | Calculate and plot allowed energy levels of an electron confined in a one-dimensional box of varying length using the particle-in-a-box model. (virtual Lab) |
| | |
| **Reference Books:** | |
| 1 | Ahluwalia, V. K., & Dhingra, S. (2017). Comprehensive practical organic chemistry. Universities Press |
| 2 | Vogel,A.I.(2012), Vogels text book of Quantitative chemical analysis, Pearson Education |
| 3 | Virtual Labs, Ministry of Education, GOI, https://vlab.co.in |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620019 | HS | -- | -- | 3 | 1.5 | 50 | 50 | 3Hrs |

**English for Technical Communication and MOODLE Laboratory**

(Common AIDS, AIML, CIC, CE, CSBS, CSD, CSE, CSIT, ECE, EEE, IT.)

**Course Objectives:**

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|---|---|
| 1 | Enable students apply correct phonological patterns for effective communication. |
| 2 | Develop students' ability to employ listening strategies to comprehend audio texts. |
| 3 | Enable students participate effectively in group discussions, presentations, debates, and interviews in professional contexts. |
| 4 | Enhance students' ability to write cohesively and coherently. |
| 5 | Familiarize students with Moodle LMS and digital learning tools for effective use of learning resources |

**Course Outcomes:**

| Sl. No | Outcome | Knowledge Level |
|---|---|---|
| 1 | Apply appropriate phonological features to communicate effectively in formal and informal contexts. | K3 |
| 2 | Apply listening strategies to comprehend audio texts, take notes, and narrate events. | K3 |
| 3 | Demonstrate effective participation in group discussions, presentations, debates, and interviews to communicate in professional contexts. | K3 |
| 4 | Develop an ability to write clearly on topics related to technology and workplace communication. | K3 |
| 5 | Demonstrate proficiency in using Moodle LMS and digital learning tools to access learning resources effectively. | K2 |

**SYLLABUS**

**PART-A**

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| 1 | Phonetics & Accent Neutralization |
| 2 | Introducing Self and Others |
| 3 | Brief Talk in Formal/Informal Contexts, JAM/Extempore |
| 4 | Listening Comprehension |
| 5 | Audio-based Note Taking & Narrating Events |
| 6 | Process Description and Paragraph Writing (Technical Lab Context) |
| 7 | Group Discussions and Deliberations |
| 8 | Micro Presentations (2-5 Mts) |
| 9 | Presenting Logical Arguments – Debates |
| 10 | Interviews – Formal & Semi-formal |

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| **PART-B**<br>**Moodle Learning Management System: Digital Learning and Academic Skills** | |
| 1 | Introduction to Moodle LMS<br>• Introduction to Moodle<br>• Student login and profile management<br>• Moodle dashboard and course navigation<br>• Accessing announcements and course materials<br>• Notifications and calendars |
| 2 | Learning Through Moodle<br>• Accessing PDFs, videos, web resources, and lecture materials<br>• Downloading and organizing study materials<br>• Moodle lessons and learning activities<br>• Tracking course completion<br>• Managing deadlines and study schedules |
| 3 | Assignments, Quizzes and Assessments<br>• Understanding assignment instructions<br>• Creating and uploading assignments<br>• File formats and file naming<br>• Online quizzes and question types<br>• Viewing marks and teacher feedback<br>• Resubmission and revision |
| 4 | Digital Communication and Collaboration<br>• Moodle forums and discussions<br>• Academic email etiquette<br>• Online communication and netiquette<br>• Group activities and peer learning<br>• Collaborative learning tools<br>• Digital presentation skills |
| | |
| **Lab Manual:** | |
| 1 | Propel: A Course Book for Technical English Communication. Cengage Publishers, 2026. |
| **Reference Books:** | |
| 1 | Richards, Jack C., Jonathan Hull, and Susan Proctor. Interchange Level 3 Student's Book with Self-study DVD-ROM. Vol. 3. Cambridge University Press, 2012. |
| 2 | Taylor Grant: English Conversation Practice,Tata McGraw Hill Education India,2016. |
| 3 | Hewing's, Martin. Cambridge Academic English (B2).CUP, 2012. |
| 4 | J.Sethi & P.V.Dhamija. A Course in Phonetics and Spoken English, (2ndEd), Kindle, 2013 |
| 5 | Raman Meenakshi,Sangeeta-Sharma.TechnicalCommunication.OxfordPress.2018. |

| **Web Resources:** | |
|---|---|
| | 1. www.esl-lab.com<br>2. www.englishmedialab.com<br>3. www.englishinteractive.net<br>4. https://www.britishcouncil.in/english/online<br>5. http://www.letstalkpodcast.com<br>6. https://www.youtube.com/c/mmmEnglish_Emma/featured<br>7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured<br>8. https://www.youtube.com/c/engvidAdam/featured<br>9. https://www.youtube.com/c/EnglishClass101/featured<br>10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists<br>11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw<br>12. https://www.youtube.com/user/letstalkaccent/videos<br>13. https://www.youtube.com/c/EngLanguageClub/featured<br>14. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc<br>15. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA |

| Course Code | Category | L | T | P | C | C.I.E. | S.E.E. | Exam |
|---|---|---|---|---|---|---|---|---|
| A2620020 | MC | 2 | -- | -- | -- | 50 | -- | -- |

**Universal Human Values – II Understanding Harmony and Ethical Human Conduct**

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

**Course Objectives:** The objective of this course is to make the student aware of

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| 1. | Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings. |
| 2. | Harmony in the human being, family, society and nature/existence |
| 3. | Holistic perspective towards life, profession and happiness |

**Course Outcomes:** At the end of this course student will be able to

| S.No | Outcome | Knowledge Level |
|---|---|---|
| 1. | **Explain** the role of value education in achieving basic human aspirations. | K2 |
| 2. | **Summarize** needs to obtain harmony in self (I). | K2 |
| 3. | **Describe** criteria for human-human relationship and harmony in society. | K2 |
| 4. | **Explain** four orders of nature and our existence. | K2 |
| 5. | **Interpret** significance of harmony in holistic development | K2 |

**SYLLABUS**

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|---|---|
| **UNIT-I (06Hrs)** | **Introduction to Value Education**:<br>Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education);<br>Understanding Value Education-Practice Session PS1 Sharing about Oneself;<br>self-exploration as the Process for Value Education;<br>Continuous Happiness and Prosperity – the Basic Human Aspirations-Practice Session PS2 Exploring Human Consciousness;<br>Happiness and Prosperity – Current Scenario;<br>Method to Fulfill the Basic Human Aspirations-Practice Session PS3 Exploring Natural Acceptance. |
| **UNIT-II (08Hrs)** | **Harmony in the Human Being**:<br>Understanding Human being as the Co-existence of the self and the body;<br>Distinguishing between the Needs of the self and the body -Practice Session PS4 Exploring the difference of Needs of self and body;<br>The body as an Instrument of the self;<br>Understanding Harmony in the self-Practice Session PS5 Exploring Sources of Imagination in the self;<br>Harmony of the self with the body;<br>Programme to ensure self-regulation and Health - Practice Session PS6 Exploring Harmony of self with the body. |
| **UNIT-III (06 Hrs)** | **Harmony in the Family and Society**:<br>Harmony in the Family – the Basic Unit of Human Interaction;<br>'Trust' – the Foundational Value in Relationship-Practice Session PS7 Exploring the |

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| | Feeling of Trust; 'Respect' – as the Right Evaluation-Practice Session PS8 Exploring the Feeling of Respect;<br>Other Feelings, Justice in Human-to-Human Relationship;<br>Understanding Harmony in the Society;<br>Vision for the Universal Human Order- Practice Session PS9 Exploring Systems to fulfil Human Goal. |
| | |
| **UNIT-IV (06 Hrs)** | **Harmony in the Nature/Existence:**<br>Understanding Harmony in the Nature;<br>Interconnectedness, self- regulation and Mutual Fulfilment among the Four Orders of Nature-Practice Session PS10 Exploring the Four Orders of Nature;<br>Realizing Existence as Co-existence at All Levels;<br>The Holistic Perception of Harmony in Existence-Practice Session PS11 Exploring Co-existence in Existence. |
| | |
| **UNIT-V (06 Hrs)** | **Implications of the Holistic Understanding – a Look at Professional Ethics**:<br>Natural Acceptance of Human Values;<br>Definitiveness of (Ethical) Human Conduct- Practice Session PS12 Exploring Ethical Human Conduct;<br>A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order;<br>Competence in Professional Ethics-Practice Session PS13 Exploring Humanistic Models in Education;<br>Holistic Technologies, Production Systems and Management Models-Typical Case Studies;<br>Strategies for Transition towards Value-based Life and Profession-Practice Session PS14 Exploring Steps of Transition towards Universal Human Order. |
| | |
| **Textbooks:** | |
| 1. | The Textbook<br>R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1 |
| 2. | The Teacher's Manual<br>R R Gaur, R Asthana, G P Bagaria,*Teachers' Manual for A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2 |
| | |
| **Reference Books:** | |
| 1. | *JeevanVidya: EkParichaya,* A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999. |
| 2. | Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004. |
| 3. | The Story of Stuff (Book). |
| 4. | The Story of My Experiments with Truth – by Mohandas Karamchand Gandhi |
| 5. | Small is Beautiful – E. F Schumacher. |
| 6. | Slow is Beautiful – Cecile Andrews |
| 7. | Economy of Permanence – J C Kumarappa |
| 8. | Bharat Mein Angreji Raj – PanditSunderlal |
| 9. | Rediscovering India – by Dharampal |
| 10. | *Hind Swaraj or Indian Home Rule* – by Mohandas K. Gandhi |
| 11. | *India Wins Freedom* – Maulana Abdul Kalam Azad |

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| 12. | *Vivekananda* – Romain Rolland (English) |
| 13. | *Gandhi* – Romain Rolland (English) |
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| **e-Resources** | |
| 1. | https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf |
| 2. | https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf |
| 3. | https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf |
| 4. | https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%2023.pdf |
| 5. | https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf |
| 6. | https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf |
| 7. | https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf |
| 8. | https://www.studocu.com/in/document/kiet-group-of-institutions/universal-human-values/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385 |
| 9. | https://onlinecourses.swayam2.ac.in/aic22_ge23/preview |